Detecting LLM-Generated Cyber Threat Intelligence Under Generator, Publisher and Rewriting Shift: A Multi-Agent Study

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***Abstract*—Fabricated cyber threat intelligence (CTI) can poison the knowledge bases and models that consume it, and large language models (LLMs) make such text cheap to produce. We test whether LLM-generated CTI can be separated from human-written reports when the generator, the publisher or the text itself changes. Human excerpts come from CISA publications released before ChatGPT, and from Microsoft and Mandiant posts kept for testing. A pipeline of three agents (Analyst, Writer, Editor) has five generators from five model families write new texts from the same extracted facts, and twelve rewriting, prompt and partial-generation conditions are derived from these texts. Detection is split across agents as well: TF-IDF models, a stylometric model, fine-tuned RoBERTa and an LLM judge feed a stacked Coordinator, with Fast-DetectGPT and Binoculars as training-free references. In-generator F1 reaches 0.998. On an unseen 120B model a TF-IDF detector falls to F1 0.329 although its ROC-AUC stays at 0.979. Training on three generators instead of one raises that F1 to 0.883, and it also raises the detector's false positive rate on Microsoft excerpts from 0.003 to 0.152. A threshold set from human scores alone shows the same trade. The error analysis explains it: the missed texts are long, impersonal and low in lexical diversity, as CISA excerpts are, and the flagged vendor excerpts are shorter, written in the first person and lexically diverse, as the training generators are. Calibration on validation data worsens the probabilities under generator shift. Paraphrasing by an LLM lowers the recall of the training-free detectors and leaves the trained ones unchanged, while rewriting from the source and sentence-level mixing lower both. The Coordinator is better than RoBERTa in 13 of 22 scenarios and worse in 5, those in which the member it weights most fails.**

***Index Terms*—cyber threat intelligence, machine-generated text detection, large language models, multi-agent systems, calibration, domain shift**

I. Introduction

CYBER threat intelligence (CTI) describes threat actors, malware, exploited vulnerabilities and the defensive steps that follow from them. Analysts read it, and machines consume it too. Indicators and narratives are parsed into knowledge graphs, turned into detection rules, and used to train or prompt other models. A fabricated report that enters this flow can therefore do harm without any human being deceived. Ranade et al. fine-tuned GPT-2 on CTI, found that threat hunters rated the generated reports as credible as real ones, and used the fake text to poison a cybersecurity knowledge graph [1]. Huang et al. fine-tuned a larger LLM on cybersecurity text and found that human annotators could not tell its CTI from genuine reports [2]. Qian et al. mixed fabricated CTI into authentic collections to simulate data poisoning [3]. Current LLMs write longer and more fluent text than these fine-tuned models, and many are free to use.

Detecting machine-generated text is an active field, but most of its evidence comes from news, essays, reviews and question answering [4]–[6]. Three results from that work matter here. Detectors trained on one generator lose accuracy on others and tend to label text from unseen generators as human [4], [5]. Paraphrasing a generated text with a second model defeats many detectors [7], [8]. Human writing that differs from the training distribution, such as non-native English, is flagged as machine-written far more often [9]. To our knowledge none of the three has been tested on CTI, and CTI differs from the benchmark domains in ways that could change them. It is dense in proper nouns and identifiers, it follows the house style of the issuing organization, and a single report is often split and re-quoted in short excerpts.

The existing CTI studies each use one fine-tuned generator and test detectors on that generator's output [1]–[3]. They do not show what happens when a detector meets a generator family it was not trained on, a larger model, a report from another organization, or a text that was rewritten after generation. They also do not show whether a failure under such a shift means that the detector has lost the signal or that its decision threshold no longer fits. The two cases need different remedies, new training data in the first and a new threshold in the second, so a study that reports only F1 at a fixed threshold cannot tell an operator what to do.

We study these questions on a corpus built for them. Human excerpts come from CISA publications released before ChatGPT, and from Microsoft and Mandiant posts of the same period that are kept out of the main training sets. LLM versions are written by a pipeline of agents. An Analyst extracts the facts of each excerpt once, and for each generator a Writer and an Editor produce a new text from those facts alone. All generators therefore describe the same content and none of them sees the human wording, which lets us read a difference between generators as a difference of writing. Detection is split across agents that look at different evidence (word and character n-grams, stylometric statistics, a fine-tuned encoder, an instruction-tuned LLM judge), and a Coordinator combines them. Two training-free detectors from the literature, Fast-

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DetectGPT [10] and Binoculars [11], serve as external reference points.

The hypothesis tested is **H1**: human-written and LLM-generated CTI differ in statistical and linguistic properties that machine learning models can use, in-generator and across generators. The paper makes four contributions.

1) We build a content-controlled CTI corpus in which five generators from five model families write from the same extracted facts. Human text from three publishers, all released on or before 30 November 2022, separates generator shift from publisher shift, and twelve rewriting, prompt-strategy and partial-generation conditions are applied to the test excerpts. All splits are made at the level of the source report.
2) We measure ranking and decision separately under shift. Three calibrators and five threshold rules are fitted on validation data only, two of them without any assumption about the test generator. The result is that both remedies for an unseen generator, pooled training data and a threshold from human scores, raise false alarms on another publisher's human text.
3) We test when combining detector agents helps. The stacked Coordinator is compared with RoBERTa, the member that holds up best under shift, in 22 scenarios with paired source-level intervals and Holm correction. Four Coordinator variants show why the combination falls behind, and inference cost is measured for every detector.
4) We analyse both kinds of error against base rates. Missed LLM texts and flagged human texts are compared with the correctly classified ones on length, lexical diversity, proper-noun share, first-person use and the n-grams the lexical model weights most, with categories, examples and regressions with source-clustered errors.

Section II reviews related work and states how this study differs from it. Sections III and IV describe the data and the agents, Section V the experimental design, and Section VI the results. Section VII discusses causes and practical use, and Section VIII the limitations.

II. Related Work

A. Supervised and zero-shot detectors

Early detectors relied on token statistics. GLTR showed that sampled text concentrates on high-rank tokens and used this to support human readers [12]. Classifiers fine-tuned on pre-trained encoders such as RoBERTa [13] detected GPT-2 output with high accuracy [14], and fine-tuned encoders remain strong baselines in shared tasks [15]. Training-free methods score a text with a reference language model. DetectGPT compares the log-probability of a passage with that of perturbed copies [16]. Fast-DetectGPT replaces the perturbation step with sampling from the model's own conditional distributions and scores what its authors call conditional probability curvature, which they report to be 340 times faster [10]. Binoculars divides the perplexity of a text under one model by the cross-perplexity between two related models, which normalizes for how surprising the prompt itself is [11]. Ghostbuster feeds token probabilities from several weaker models into a linear classifier and generalizes well across domains [17]. Its feature models are OpenAI models that are no longer served, so we could not reproduce it.

B. Generalization, attacks and partial generation

M4 covers several generators, domains and languages, and reports that detectors often label text from unseen generators as human [4]. RAID shows that unseen generators, decoding settings and adversarial edits defeat most detectors [5], and DetectRL builds its test conditions from the ways generated text is used in practice, including paraphrasing and mixing human with machine text [6]. Krishna et al. show that a paraphrasing model evades several detectors [7], and Sadasivan et al. relate the best achievable AUROC to the distance between the human and machine distributions [8]. RADAR trains a detector against a paraphraser to make it harder to evade by rewriting [18]. When only part of a text is generated, the task becomes locating the boundary between human and machine sentences, as in SemEval-2024 Task 8 [15], [19]. Detectors also misfire on human text that differs from their training data: non-native English writing is flagged far more often than native writing [9].

C. Ensembles and multi-agent detection

Stacked generalization trains a second-level model on the outputs of first-level models [20]. Ong and Quek combine the outputs of several DetectGPT classifiers and report a higher AUROC than DetectGPT alone when the generator is unknown [21]. Two recent systems assign detection to cooperating LLM agents. In CAMF, three agents profile the style, the semantic coherence and the reasoning of a text, two further agents argue over inconsistencies between the profiles, and a judge decides [22]. AGENT-X routes a text to expert agents that apply linguistic guidelines and needs no decision threshold [23]. Both are built from prompted LLMs without training and are evaluated on general-domain benchmarks (news, essays, stories, question answering) with text from GPT-3.5, GPT-4 and Claude models. They therefore say little about a learned combination of trained detectors, and in particular about whether such a combination keeps its advantage over its members when the generator changes.

D. Calibration under shift

Modern neural classifiers are often over-confident, and temperature scaling is a simple fix in-distribution [24]. Platt scaling [25], isotonic regression [26] and temperature scaling are fitted on held-out data from the training distribution. Under dataset shift, calibration from in-distribution data degrades and can give little protection [27]. Calibration error is commonly measured with binned expected calibration error [28], the Brier score and the negative log-likelihood. For text detection the practical case is a detector whose ROC-AUC stays high on a new generator while its F1 falls. We ask whether the threshold is then the problem and whether it can be repaired without data from that generator.

E. LLMs and CTI

Ranade et al. generated fake CTI with fine-tuned GPT-2 for data poisoning [1]. Huang et al. fine-tuned an LLM on domain data, showed that annotators could not separate its CTI from genuine reports, and benchmarked misinformation detectors on it [2]. Qian et al. fine-tuned a general language model to generate fake CTI and proposed a BERT and TextCNN classifier to find it [3].

F. How this study differs

The CTI studies above use one fine-tuned generator, one source of human text and a fixed decision threshold. This study differs in four respects. (i) Generation is content-controlled: five generators from five families write from identical extracted facts and none sees the human wording, so cross-generator results are not confounded by topic. (ii) Shift is tested along three separate axes, generator (unseen families and larger models), publisher (Microsoft and Mandiant human text never used in the main training sets) and rewriting (paraphrase, back-translation, a "humanize" prompt, prompt strategies and sentence-level mixing), with paired source-level statistics. (iii) Calibration and threshold choice are studied as a question separate from ranking, with every fit restricted to training and validation data. (iv) The multi-agent detector is not assumed to help. It is compared with its strongest member under shift scenario by scenario, next to two published zero-shot detectors and a cost measurement. The general-domain benchmarks [4]–[6] cover more generators and attacks than we do. They do not cover the CTI register, in which identifiers, agency names and house style carry much of the signal.

III. Data

A. Human-written CTI

CISA (training, validation and test). We collected CISA cybersecurity advisories, analysis reports and alerts, which are in the public domain. Release dates were read from the advisory code or URL, because the pages show only the last revision date. We kept documents released on or before 30 November 2022, before ChatGPT was public, so that human authorship is not in doubt. Vendor patch notices and catalog updates were excluded. From each page we kept narrative <p> paragraphs of at least 20 words, dropped tables, code, indicator lists and boilerplate, and joined consecutive paragraphs into excerpts of 150–350 whitespace-separated words, at most 3 per source. This gave 203 sources and 406 excerpts. Each source received a topic label (APT, ransomware, phishing, vulnerability, malware, other) from keyword rules. Word counts in the tables are the alphabetic tokens counted by the stylometric features, which are slightly fewer.

Microsoft and Mandiant (independent test only). To test whether the detectors confuse another organization's writing style with LLM authorship, we collected two vendor sources with the same date cutoff: the Microsoft Security Blog and the Mandiant threat intelligence blog, now hosted by Google Cloud. For Microsoft we kept posts that the blog itself labels as content type *Research* and topic *Threat intelligence* and excluded product news and best-practice posts. Microsoft's page metadata show modification dates from a recent site migration, not a revision history, so we dated posts by the publication date in the URL and removed paragraphs that announce later updates (for example, renamed threat actors or "2023 update" notes). Mandiant posts carry their original date only in the visible byline. The Mandiant feed also contains tooling and red-team posts, which we excluded by title rules. Excerpts were built as for CISA. We sampled 80 Microsoft and 80 Mandiant sources (231 and 207 excerpts) with a fixed seed. Table I gives counts, date ranges and lengths. These texts are never part of the main training sets. One augmented experiment adds one publisher to training and tests on the other (Section V). The pages are copyrighted, so the repository holds their URLs and excerpt hashes and the collection script rebuilds the text.

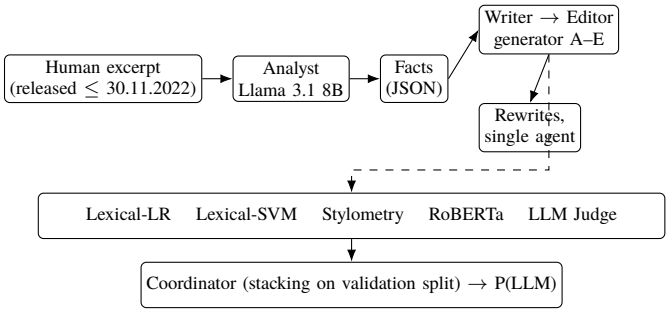


Fig. 1. Generation agents (top) and detection agents (bottom). All generators write from the same Analyst facts and never see the human excerpt. Rewritten versions start from the Writer–Editor output; the single-agent condition reads the human excerpt directly. Human excerpts come from CISA (training and testing) and from Microsoft and Mandiant (testing only).

B. Multi-agent generation

Fig. 1 summarizes the pipeline. Each agent is a role prompt bound to one model. It receives one message and returns one. The *Analyst* reads a human excerpt and returns the facts it states as JSON with ten keys (actor, malware, CVEs, TTPs, targets, sectors, indicator types, timeline, impact, mitigations), with the instruction to leave a key empty when the excerpt says nothing about it. It runs once per excerpt on a local Llama 3.1 8B with JSON-constrained output at temperature 0.2. For each generator, a *Writer* receives the facts and a target length equal to that of the human excerpt and writes a CTI text. An *Editor* then receives the facts and the draft and revises the draft against the facts, with the instruction not to add new ones. The Writer prompt is one of four templates (SOC bulletin, vendor blog, government advisory, incident report) and the sampling temperature is 0.7 or 1.0. Both are drawn from a random generator seeded by generator and excerpt, so an interrupted run resumes with the same choices, and both are stored with every text together with all agent messages and the model name. A reply that is empty or opens with a refusal is requested again, up to three times, after which the excerpt is skipped for that generator.

The training generators run locally through Ollama: A, Llama 3.1 8B (Meta); B, Qwen 2.5 7B (Alibaba); and C,

TABLE I
HUMAN-WRITTEN CTI BY PUBLISHER. ALL EXCERPTS ARE FROM DOCUMENTS RELEASED ON OR BEFORE 30 NOVEMBER 2022.

Publisher	Sources	Excerpts	Released	Median words	IQR words	Role
CISA	203	406	2015-02-20–2022-11-17	302	239–326	Train, val., test
Microsoft	80	231	2012-01-16–2022-11-22	319	298–334	Independent test
Mandiant	80	207	2010-08-16–2022-11-28	306	280–322	Independent test

Mistral 7B (Mistral AI). Two larger API models serve only as unseen test generators: D, gpt-oss-120b (OpenAI weights, Groq API), and E, Gemini 3.1 Flash-Lite (Google API). Their free tiers allow about 200k tokens and 500 requests per day. After an initial unrestricted run we therefore restricted D and E to excerpts whose source falls in the test split of at least one seed, and to the multi-agent condition. All 191 such excerpts have a D and an E version. Texts generated before the restriction bring the coverage to 61% and 75% of all excerpts. They are never used for training and enter only the corpus statistics. For a fixed 20% of the sources, each local generator also produces a *single-agent* version, one call that reads the human excerpt and rewrites it. The external excerpts receive LLM versions from A, B and C through the same pipeline.

C. Rewriting, prompt strategies and partial generation

For every excerpt whose source is in the test split of at least one seed, we derived further LLM texts from the multi-agent texts of A, B and C (Table VI lists them). Five conditions rewrite a finished text: *paraphrase, same model* (the generator rewrites its own text with an instruction to keep every fact, name and number), *paraphrase, other model* (A's text rewritten by B, B's by C, C's by A), *paraphrase twice* (the same-model paraphrase paraphrased again), *back-translation* through German, and a *"humanize"* prompt that asks for the register of an experienced analyst at a national CERT, without first person and without advice to the reader. Two prompt strategies change the Writer: an *advisory-style* prompt that asks for the impersonal register of a government advisory, and a *few-shot* prompt that shows two human excerpts taken from sources that are in no seed's test split. The *Writer-only* condition is the Writer's draft before the Editor, which measures what the third agent changes. Finally, *partly generated* texts join sentences of a human excerpt with sentences of its own LLM version so that LLM sentences make up about 25%, 50% or 75% of the words. The LLM block is placed first or last at random, because RoBERTa reads only the first 256 tokens. In total this adds 6293 LLM texts. We label every partly generated text as LLM. This is a convention: a text with a quarter of LLM words is mostly human, and "recall" on these texts measures how much inserted LLM text a detector needs before it responds.

We did not build a human-edited set, in which people revise LLM drafts. Simulating such edits with another LLM would measure LLM rewriting a second time, so we leave this condition to future work and use the sentence-level mixtures as the closest available proxy (Section VIII).

D. Preprocessing and splits

LLM outputs contain wrapping that is not part of the text: reasoning tags, preambles such as "Here is the revised text", titles, markdown, bullet markers and word-count notes. We remove these and normalize Unicode variants of dashes and quotes in all texts, human and generated. The effect of this step is measured in an ablation. Texts under 50 words are dropped.

Splits are made by source report: 70% of CISA sources for training, 10% for validation and 20% for testing, separately for each seed (42, 123, 2024). A human excerpt and all of its LLM versions, rewritten or not, always fall in the same split. Vectorizers, scalers, detectors, the Coordinator, calibrators and thresholds are fitted on training or validation data only. Test, D, E, rewritten and external texts are only scored.

IV. DETECTION AGENTS

A. Members

- **Lexical-LR** (baseline 1): word 1–2-gram and character 3–5-gram TF-IDF with logistic regression.
- **Lexical-SVM** (baseline 2): the same features with a linear SVM, Platt-calibrated.
- **Stylometry**: 32 features in five groups (length, lexical diversity including MTLD [29], punctuation, repetition, part-of-speech shares) with XGBoost.
- **Transformer**: RoBERTa-base [13] fine-tuned for three epochs, maximum length 256 tokens.
- **LLM Judge**: Gemma 2 9B, a model family not used for generation, asked zero-shot for the probability that a text is AI-generated.

B. Training-free reference detectors

Fast-DetectGPT uses the analytic form of the conditional probability curvature with GPT-Neo 2.7B as both the sampling and the scoring model [10]. *Binoculars* uses the authors' pair, Falcon-7B as observer and Falcon-7B-Instruct as performer [11]. Two 7B models do not fit together on an 8 GB GPU, so both are loaded in 4-bit precision and run in two passes. The observer's next-token distribution is stored on disk, truncated to its 64 most probable tokens and renormalized, and the performer pass computes perplexity and cross-perplexity from it. Without truncation this two-pass computation reproduces the exact score, which we checked on a small model. The truncation is not harmless. On 16 randomly chosen texts scored both ways, the truncated score was higher than the exact one in 16 cases, with a median absolute difference of 1.5×10^{-1} against a median exact score of 0.930, while the rank correlation between the two was 0.944. Our Binoculars scores are therefore shifted but order the texts almost as the exact ones do. Both methods read at most 512 tokens. They need no training, but their scores are not probabilities. We map them to probabilities with class-balanced Platt scaling on the

validation split of each training set, so that they are compared with the supervised detectors under the same threshold rules.

C. Coordinator

The Coordinator is a logistic regression on the logits of the five members' probabilities (stacked generalization [20]),

$$P(\text{LLM} \mid x) = \sigma\Big(b + \sum_{m=1}^{5} w_m \log \frac{p_m(x)}{1 - p_m(x)}\Big), \qquad (1)$$

where $p_m(x)$ is the probability from member m and σ the logistic function. The weights w_m and the bias b are fitted on the validation split of the training generators, which the members have not been trained on. They are logged per run and reported with the results. To find out why a stack can fall behind its best member, we evaluate four variants on the same predictions: the unweighted mean of member logits (no learned weights), the single member with the best validation F1 (selection in place of combination), a stack fitted on validation texts of local generators that are *not* in the training set (weights learned under a simulated generator shift), and the stack with the two zero-shot detectors added as members.

D. Why several agents

Generation. Splitting fact extraction from writing serves the experiment. When one prompt reads the human excerpt and rewrites it, as in our single-agent condition, the output keeps names and phrasing of the source, and different generators keep different amounts of it. Between 0.237 and 0.359 of the word trigrams of a single-agent text also occur in its human excerpt, depending on the generator, against 0.091 to 0.145 for multi-agent texts of the same excerpts, where the remaining overlap comes from entity names and stock phrases that pass through the facts. Cross-generator differences in the single-agent condition would thus mix style with copying. With a shared Analyst, the content is fixed across all five generators and the human wording is never shown to the Writer. The Editor adds a review step, as in human CTI production, and its effect is measured by the Writer-only condition. The price is three calls per text instead of one.

Detection. The members use different evidence: surface n-grams, distributional statistics of style, contextual representations, the judgment of an instruction-tuned model and, for the reference detectors, likelihood under another language model. A generator shift does not affect these equally. A larger generator may match human length, which removes a stylometric cue, while keeping the lower proper-noun share that the lexical models use. If member errors are not perfectly correlated, a learned combination can be more accurate than any member. The same argument says where it should fail. Weights fitted on validation texts of the training generators reward the member that is most reliable there, and that member need not be the one that transfers best. The variants above test this directly, and Section VI reports when the combination helps and when it does not.

V. Experimental Design

A. Scenarios

One training run is made for every non-empty subset of the local generators (A, B, C, AB, AC, BC, ABC) and every seed, and each trained model is tested on the test split of every generator. This yields four settings from the same runs: *E1 in-generator* (train and test on the same generator, plus the pooled ABC model), *E2 cross-generator* (train on one, test on another; the required Human+A $\rightarrow$ Human+B and Human+A $\rightarrow$ Human+C cases are part of it), *E3 leave-one-out* (train on two, test on the third) and *E4 external generator* (any training set, tested on D or E). Each test set holds the LLM texts of one generator for the test sources and the human excerpts they were made from, so the classes are balanced.

The same runs score the external publishers and the rewritten texts. *Publisher shift* is measured by the false positive rate on all Microsoft and Mandiant human excerpts and by recall, F1 and ROC-AUC on those excerpts paired with their A, B and C versions. *Rewriting and partial generation* are measured by recall on the derived texts, which is compared with recall on the multi-agent texts of the same excerpts. The single-agent condition is compared with multi-agent texts of the same model and the same excerpts. Two augmented trainings, with three seeds each, test whether more varied training data help: ABC plus one external publisher (human and LLM versions), tested on the other publisher, and ABC plus paraphrased and single-agent texts of the training sources.

B. Metrics and statistics

We report accuracy, precision, recall, F1, macro-F1, ROC-AUC and PR-AUC as mean $\pm$ standard deviation over three seeds. Intervals and tests work at the level of source reports. Test sources are resampled with replacement (2,000 bootstrap samples), which gives a 95% percentile interval for a metric, and for a comparison the metric difference is recomputed on the same resampled sources for both systems, which gives a paired interval and a two-sided p-value. With 2,000 samples the smallest such p-value is $2/2001$, so Holm's correction [30] cannot reject anything in a family of more than fifty tests. We therefore pool the runs of a comparison before testing: one test per detector and target for the in-generator against cross-generator gap, one per scenario for the Coordinator against RoBERTa, one per condition and detector for the rewriting conditions. Holm's method is applied within each of these families. Where tests are reported per run or per feature, the Benjamini-Hochberg procedure [31] is used. Detectors are also compared on single test sets with exact McNemar tests, Holm-corrected per detector pair. Linguistic differences are tested with Mann-Whitney U and Cliff's δ as the effect size, and distinctive n-grams are ranked with weighted log-odds [32].

C. Calibration and thresholds

For each detector and training run we fit three calibrators on the validation split: Platt scaling (logistic regression on the logit of the score) [25], temperature scaling (a single divisor of the logit chosen by validation log-loss) [24] and isotonic

regression [26]. Because the Coordinator is itself fitted on the validation split, its calibrators use out-of-fold stacked scores from a five-fold split grouped by source. Probability quality is measured on the test sets with the Brier score, the expected calibration error with 15 equal-width bins [28] and the negative log-likelihood, and shown in reliability diagrams (Fig. 7).

Five decision rules are compared, all set on validation data: the fixed threshold 0.5; the threshold with the best validation F1; a *pseudo-unseen* threshold with the best F1 on validation texts of local generators that are neither in the training set nor the test generator; and two *human-quantile* thresholds, the 99th and 95th percentiles of validation human scores, which aim at a 1% or 5% false positive rate and use no LLM text at all. The last two rules assume nothing about the generator met at test time. They do assume that human text at test time scores like CISA validation text, and the external publishers test that assumption.

D. Cost

Inference time is measured on one laptop RTX 4060 (8 GB) for a fixed sample of test texts, with three repetitions and nothing else running on the GPU. The LLM judge is timed on a smaller sample. Training time is logged in every run. The Coordinator's cost is the sum of its members run one after another.

E. Error analysis

False negatives (LLM texts scored as human) are taken from models trained on one local generator, over all seeds, because that is where D is missed. False positives (human texts scored as LLM) are taken from the same models for CISA and from the pooled ABC models for the vendor excerpts, because that is where the false alarms occur. For every text we compute length, MTLD, type-token ratio, proper-noun and pronoun shares, first-person pronouns (*we, our, us, ours*), agency names, advice terms, technical identifiers (file names, CVE numbers, addresses, ATT&CK identifiers) that the Analyst facts do not contain, and the rate of *cues*: occurrences per 100 words of the word n-grams among the 30 terms with the largest positive and the 30 with the largest negative weight in Lexical-LR trained on A (23 and 20 of them are word n-grams).

Three analyses use these properties. First, each error is assigned to one category by the first rule that applies, and the share of every category among the errors is reported next to its share among all texts of the same group. Without the second number a category that merely describes the generator would look like a cause. Second, misclassified texts are compared with correctly classified ones property by property (Mann-Whitney U, Cliff's δ, Benjamini-Hochberg), after averaging each text over runs so that a text missed in every run is not counted nine times. Third, a logistic regression of the error indicator on standardized properties, with standard errors clustered by source, estimates which properties change the odds of an error when the others are held fixed. One author defined the rules, and they were not validated by a second annotator.

VI. Results

A. Linguistic differences

Table II and Fig. 2 summarize the corpus. Of the 160 generator-feature comparisons, 61 show a large effect ($|\delta| \geq 0.474$) after Benjamini-Hochberg correction. Lexical diversity differs most consistently. The median MTLD is 84 for the CISA excerpts and between 90 and 199 for the generators, and the hapax ratio is higher for every generator (δ from 0.603 to 0.932). LLM texts repeat fewer bigrams and use fewer proper nouns. The human excerpts name agencies, products, files and dates again and again, while the generators restate the same facts in general terms. The local generators also write short texts (median 163, 147 and 214 words against 302), whereas D and E match the human length (306 and 300). The log-odds ranking points the same way. LLM texts over-use *our*, *we*, *potential*, *to mitigate* and *defenders should*, the vocabulary of advice addressed to a reader. The CISA excerpts over-use *CISA*, *FBI*, *see* and *note*, the vocabulary of attribution and cross-reference. H1 holds for all five generators at the level of single features.

The two vendor publishers sit between CISA and the generators on some of these features. Against CISA, Microsoft and Mandiant excerpts use more pronouns ($\delta = 0.785$ and 0.648) and fewer proper nouns ($\delta = -0.595$ and -0.430), the direction in which LLM text differs from CISA. 11 of 64 publisher-feature comparisons show a large effect.

B. In-generator and cross-generator detection

Table III gives the main result for models trained on one generator. In-generator detection is close to perfect for the lexical baselines (F1 0.997), RoBERTa (0.992) and the Coordinator (0.998). Moving to another local generator costs little. Lexical-LR falls to 0.966 and RoBERTa to 0.990, and leave-one-out training on two generators gives similar values (Lexical-LR 0.997, RoBERTa 0.984). Gemini (E) is detected almost as well by these detectors. The large drop is on gpt-oss-120b (D). Lexical-LR falls to 0.329 (95% CI [0.288, 0.367]) and Lexical-SVM to 0.327, while RoBERTa keeps 0.859 ([0.834, 0.880]) and the Coordinator 0.721 ([0.689, 0.751]). Fig. 3 shows every training set.

Lexical-LR still ranks the D texts above the human ones (AUC 0.979), but at the fixed threshold it finds only 0.208 of them. In Fig. 4 the D scores form a band of their own between the human excerpts and the local generators, mostly below 0.5, so a lower threshold would separate them from the human texts. Training on more generators moves the boundary. With two generators Lexical-LR reaches F1 0.764 on D, and with all three 0.883. RoBERTa reaches 0.958 and the Coordinator 0.915 with three.

The two training-free detectors are well below the supervised ones even in-generator (AUC 0.828 for Fast-DetectGPT and 0.900 for Binoculars), and their ranking degrades on the larger generators (AUC on D 0.699 and 0.734, on E 0.613 and 0.671). The LLM Judge is close to chance (F1 0.504). Its in-generator AUC of 0.409 is below 0.5 because it rates 82% of the CISA test excerpts as AI-written, more than it does for the texts of some generators.

TABLE II
CORPUS PER SOURCE. MEDIANS OVER MULTI-AGENT TEXTS (HUMAN EXCERPTS FOR THE FIRST ROW).

Source	Multi	Single	Words	MTLD	Rep. bigram	PROPN
Human (CISA)	406	0	302	84	0.094	0.114
A: Llama 3.1 8B	406	83	163	90	0.065	0.050
B: Qwen 2.5 7B	405	83	147	121	0.033	0.055
C: Mistral 7B	406	83	214	132	0.039	0.059
D: gpt-oss-120b	248	0	306	199	0.029	0.049
E: Gemini 3.1 Flash-Lite	305	0	300	169	0.031	0.033

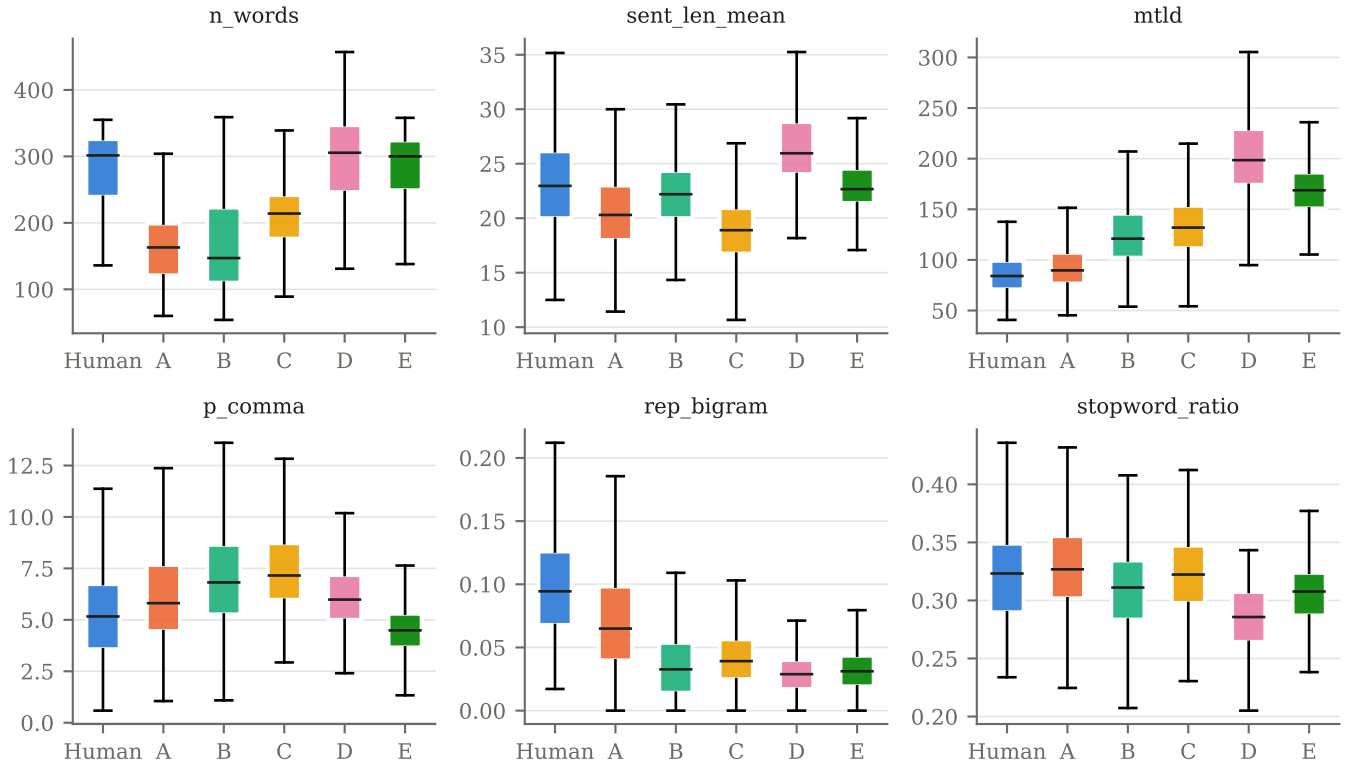


Fig. 2. Selected stylometric features for CISA excerpts and the five generators (multi-agent texts).

Pooled over runs, the F1 lost by moving from the training generator to another local generator is 0.030 for Lexical-LR (95% CI [0.023, 0.038]) and 0.002 for RoBERTa ([0.000, 0.005]). After Holm correction the loss is significant for 5 of the 7 trained and training-free detectors, RoBERTa and Binoculars being the exceptions. On D the loss is 0.667 for Lexical-LR, 0.134 for RoBERTa and 0.277 for the Coordinator, and it is significant for all 7 detectors. On E it is significant for 5 of 7, and neither RoBERTa nor the Coordinator loses measurably. Tested run by run, 37 of 126 local comparisons and 86 of 126 comparisons on D and E remain significant after Benjamini-Hochberg correction. Exact McNemar tests on single test sets agree with this picture. After Holm correction RoBERTa differs from Lexical-LR on 20 of 105 test sets, the Coordinator from Lexical-LR on 17 and the Coordinator from RoBERTa on 5.

Table IV reports every metric for the required setting, training on Human + A, and Fig. 5 shows the Coordinator's confusion matrices for seed 42. On A it makes 1 false positive and 0 false negatives (TP 74, TN 73). On D it still makes 1 false positive but misses 44 of 74 LLM texts. In a CTI pipeline the false negative is the costly error. A missed fabricated report enters the knowledge base and is shared onward as intelligence, while a false positive costs an analyst a second look.

C. Publisher shift

The models that handle D best raise false alarms on vendor writing (Table V). Trained on CISA and the three local generators, Lexical-LR flags 0.152 of Microsoft and 0.137 of Mandiant excerpts as LLM-written, against 0.014 of CISA test excerpts. For RoBERTa the rates are 0.159 and 0.198 against 0.042. The Coordinator is lower (0.046 and 0.064). On vendor excerpts paired with their own LLM versions all three still reach an AUC of 1.000, so the vendor human texts are scored above the threshold but below their LLM versions.

The false alarms grow with the number of generators in training. Lexical-LR trained on one generator flags 0.003 of Microsoft and 0.003 of Mandiant excerpts, on two generators

TABLE III
MODELS TRAINED ON ONE GENERATOR (A, B OR C). F1 IS MEAN ± STD OVER THREE SEEDS OF THE AVERAGE OVER TRAINING GENERATORS; AUC IS ROC-AUC.

Detector	In-generator		Cross (A–C)		D: gpt-oss-120b		E: Gemini	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC
Lexical-LR	0.997 ± 0.004	1.000	0.966 ± 0.008	1.000	0.329 ± 0.017	0.979	0.961 ± 0.012	1.000
Lexical-SVM	0.998 ± 0.004	1.000	0.976 ± 0.004	1.000	0.327 ± 0.013	0.983	0.969 ± 0.011	1.000
Stylometry	0.934 ± 0.006	0.984	0.868 ± 0.004	0.959	0.706 ± 0.013	0.878	0.778 ± 0.022	0.931
RoBERTa	0.992 ± 0.003	1.000	0.990 ± 0.002	1.000	0.859 ± 0.022	0.982	0.991 ± 0.003	1.000
LLM Judge	0.504 ± 0.024	0.409	0.504 ± 0.024	0.409	0.499 ± 0.026	0.533	0.628 ± 0.015	0.754
Fast-DetectGPT	0.744 ± 0.009	0.828	0.714 ± 0.033	0.828	0.540 ± 0.054	0.699	0.427 ± 0.053	0.613
Binoculars	0.817 ± 0.024	0.900	0.806 ± 0.025	0.900	0.365 ± 0.042	0.734	0.307 ± 0.060	0.671
Coordinator	0.998 ± 0.000	1.000	0.993 ± 0.003	1.000	0.721 ± 0.083	0.987	0.995 ± 0.003	1.000

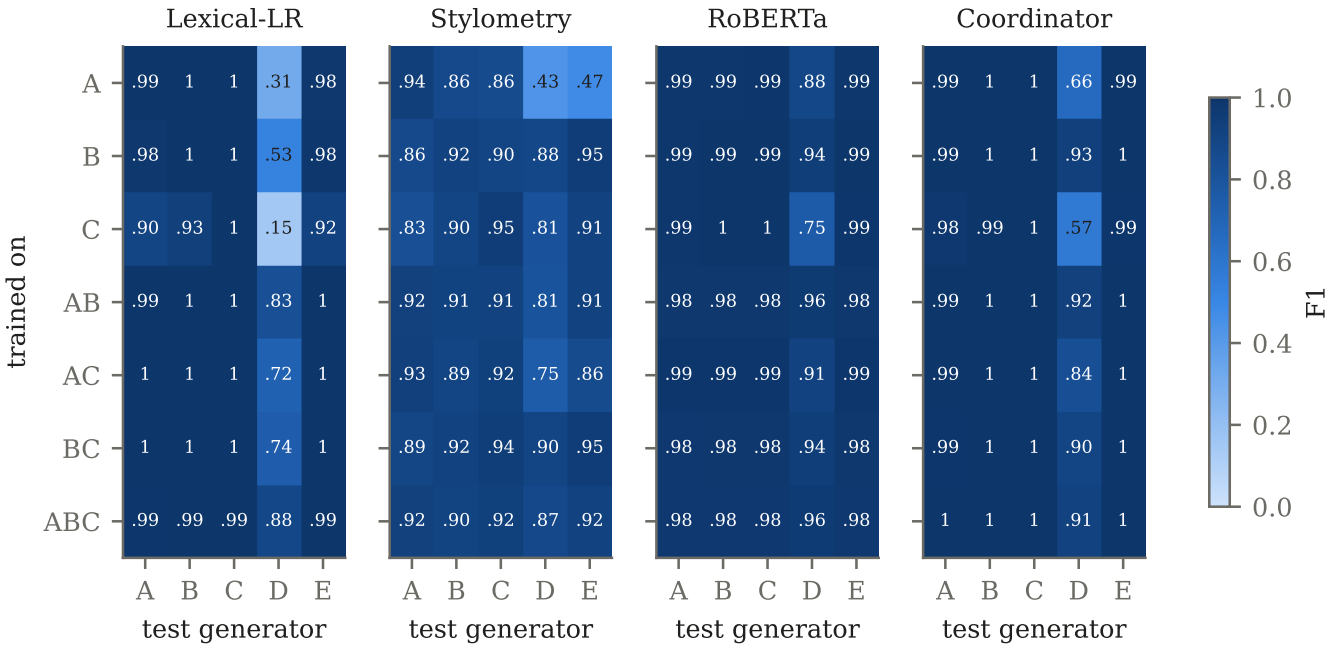


Fig. 3. F1 for every training set (rows) and test generator (columns), mean over three seeds. D and E are never used for training.

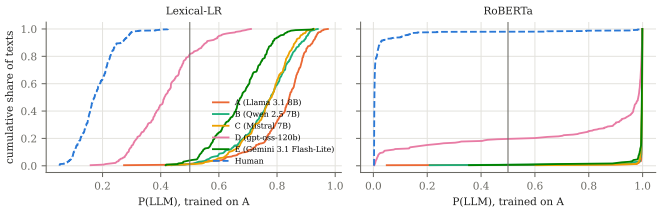


Fig. 4. Cumulative distribution of P(LLM) for models trained on A. Human excerpts (dashed) and each generator's texts; the vertical line is the fixed threshold.

0.041 and 0.053, and on three the values above. RoBERTa goes from 0.075 and 0.093 to 0.140 and 0.159 with two. The pooling that recovers D in the previous subsection is therefore the same step that produces the vendor false alarms. A model that has seen three ways of writing LLM text draws a wider region around "LLM", and human text that is unlike CISA falls into it.

Adding one vendor publisher to training removes most of the problem on the other. The false positive rate on Microsoft falls to 0.016 (Lexical-LR), 0.022 (RoBERTa) and 0.000 (Coordinator), and on Mandiant to 0.011, 0.052 and 0.023. The training-free detectors behave the other way round. Their false positive rate is lower on vendor text than on CISA text (Binoculars 0.030 and 0.016 against 0.093; Fast-DetectGPT 0.110 and 0.116 against 0.222). A scoring language model has no notion of CISA's style, so vendor prose does not look unusual to it. Section VI-I examines which vendor excerpts are flagged.

D. Rewriting, prompt strategy and partial generation

Rewriting an LLM text with an LLM does not hide it from the trained detectors (Table VI). Paraphrasing by the same model, by another model or twice, back-translation and the "humanize" prompt all leave the recall of Lexical-LR, RoBERTa and the Coordinator at 0.992 or above. The Writer-only drafts are detected as well as the edited texts, so the Editor does not change what these detectors respond to. The few-shot prompt, which shows two human excerpts to the Writer, lowers recall slightly (Lexical-LR 0.982, RoBERTa 0.969,

TABLE IV
TRAINING ON HUMAN + A, TESTING ON HUMAN + EACH GENERATOR (MEAN ± STD OVER SEEDS).

Test	Detector	Acc	Precision	Recall	F1	Macro-F1	ROC-AUC	PR-AUC
A	Lexical-LR	0.994 ± 0.006	1.000 ± 0.000	0.988 ± 0.013	0.994 ± 0.006	0.994 ± 0.006	1.000 ± 0.000	1.000 ± 0.000
A	RoBERTa	0.987 ± 0.014	0.979 ± 0.027	0.996 ± 0.007	0.987 ± 0.013	0.987 ± 0.014	1.000 ± 0.000	1.000 ± 0.000
A	Coordinator	0.994 ± 0.000	0.992 ± 0.007	0.996 ± 0.007	0.994 ± 0.000	0.994 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
B	Lexical-LR	0.996 ± 0.004	1.000 ± 0.000	0.991 ± 0.008	0.996 ± 0.004	0.996 ± 0.004	1.000 ± 0.000	1.000 ± 0.000
B	RoBERTa	0.987 ± 0.012	0.979 ± 0.027	0.996 ± 0.007	0.987 ± 0.012	0.987 ± 0.012	1.000 ± 0.001	1.000 ± 0.001
B	Coordinator	0.996 ± 0.004	0.992 ± 0.007	1.000 ± 0.000	0.996 ± 0.004	0.996 ± 0.004	1.000 ± 0.000	1.000 ± 0.000
C	Lexical-LR	0.996 ± 0.004	1.000 ± 0.000	0.991 ± 0.008	0.996 ± 0.004	0.996 ± 0.004	1.000 ± 0.000	1.000 ± 0.000
C	RoBERTa	0.989 ± 0.014	0.979 ± 0.027	1.000 ± 0.000	0.989 ± 0.014	0.989 ± 0.014	1.000 ± 0.001	1.000 ± 0.001
C	Coordinator	0.996 ± 0.004	0.992 ± 0.007	1.000 ± 0.000	0.996 ± 0.004	0.996 ± 0.004	1.000 ± 0.000	1.000 ± 0.000
D	Lexical-LR	0.593 ± 0.004	1.000 ± 0.000	0.185 ± 0.009	0.313 ± 0.012	0.512 ± 0.007	0.980 ± 0.018	0.980 ± 0.019
D	RoBERTa	0.894 ± 0.065	0.977 ± 0.029	0.810 ± 0.149	0.879 ± 0.087	0.892 ± 0.069	0.977 ± 0.017	0.977 ± 0.018
D	Coordinator	0.747 ± 0.047	0.983 ± 0.016	0.503 ± 0.093	0.662 ± 0.084	0.730 ± 0.057	0.972 ± 0.014	0.972 ± 0.016
E	Lexical-LR	0.981 ± 0.001	1.000 ± 0.000	0.962 ± 0.002	0.981 ± 0.001	0.981 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
E	RoBERTa	0.987 ± 0.012	0.979 ± 0.027	0.996 ± 0.007	0.987 ± 0.012	0.987 ± 0.012	1.000 ± 0.001	1.000 ± 0.001
E	Coordinator	0.994 ± 0.000	0.992 ± 0.007	0.996 ± 0.007	0.994 ± 0.000	0.994 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

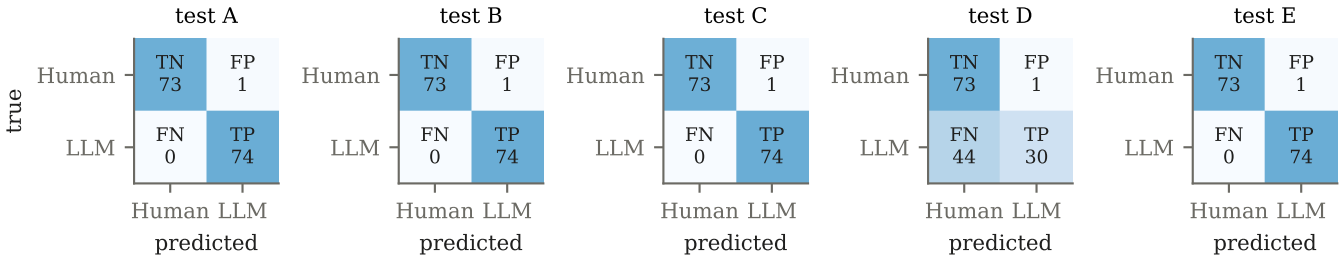


Fig. 5. Coordinator trained on Human + A (seed 42), tested on each generator.

TABLE V
MODELS TRAINED ON CISA + ABC, THRESHOLD 0.5. FPR ON ALL HUMAN EXCERPTS OF THE PUBLISHER; RECALL, F1 AND AUC ON HUMAN EXCERPTS PAIRED WITH THEIR LLM VERSIONS FROM A, B AND C, AVERAGED. FPR AND F1 ARE MEAN ± STD OVER THREE SEEDS; THE INTERVAL RESAMPLES SOURCES. THE LAST COLUMN ADDS THE OTHER EXTERNAL PUBLISHER TO TRAINING.

Publisher	Detector	FPR	95% CI	Recall	F1	ROC-AUC	FPR, other publisher in training
CISA	Lexical-LR	0.014±0.023	–	0.999	0.993±0.011	1.000	–
CISA	RoBERTa	0.042±0.007	–	1.000	0.979±0.003	1.000	–
CISA	Coordinator	0.009±0.007	–	1.000	0.996±0.004	1.000	–
CISA	Fast-DetectGPT	0.222±0.048	–	0.721	0.739±0.029	0.828	–
CISA	Binoculars	0.093±0.008	–	0.750	0.811±0.031	0.900	–
Microsoft	Lexical-LR	0.152±0.009	[0.126, 0.177]	1.000	0.930±0.004	1.000	0.016±0.007
Microsoft	RoBERTa	0.159±0.029	[0.134, 0.184]	1.000	0.927±0.012	1.000	0.022±0.004
Microsoft	Coordinator	0.046±0.046	[0.033, 0.061]	1.000	0.978±0.022	1.000	0.000±0.000
Microsoft	Fast-DetectGPT	0.110±0.025	[0.086, 0.134]	0.874	0.880±0.008	0.954	–
Microsoft	Binoculars	0.030±0.007	[0.017, 0.045]	0.885	0.924±0.008	0.979	–
Mandiant	Lexical-LR	0.137±0.012	[0.111, 0.166]	1.000	0.936±0.005	1.000	0.011±0.003
Mandiant	RoBERTa	0.198±0.041	[0.162, 0.237]	1.000	0.910±0.017	1.000	0.052±0.023
Mandiant	Coordinator	0.064±0.016	[0.045, 0.086]	1.000	0.969±0.007	1.000	0.023±0.007
Mandiant	Fast-DetectGPT	0.116±0.017	[0.086, 0.150]	0.825	0.848±0.011	0.940	–
Mandiant	Binoculars	0.016±0.006	[0.003, 0.031]	0.826	0.895±0.008	0.977	–

Coordinator 0.950). The training-free detectors are affected. Paraphrasing lowers the recall of Fast-DetectGPT from 0.721 to 0.572 and that of Binoculars from 0.749 to 0.548, and back-translation lowers Binoculars to 0.495, as reported for likelihood-based detectors on general text [7]. Both also detect the Writer's unedited drafts better than the final texts (0.839 and 0.855), so the Editor pass makes the output less predictable to a scoring model.

Single-agent rewriting, in which the model reads the human excerpt and rewrites it, does hide LLM text. Recall falls to 0.663 (Lexical-LR), 0.721 (RoBERTa) and 0.601 (Coordinator), against 1.000 on multi-agent texts of the same model and excerpts. The test sets are small, 138 single-agent texts over the three generators and seeds, which shows in the standard deviations of Table VI. These are the texts that share a quarter to a third of their word trigrams with the source (Section IV-D).

Partly generated texts are the hardest case (Fig. 6). With a quarter of the words from the LLM, recall is 0.303, 0.366 and 0.171, and with half of them 0.844, 0.620 and 0.535. For RoBERTa the position of the LLM block decides. At 50% it finds 0.947 of the texts that begin with the LLM block

TABLE VI

RECALL ON LLM TEXTS AT THRESHOLD 0.5 UNDER REWRITING, PROMPT STRATEGY AND PARTIAL GENERATION, TRAINED ON CISA + ABC (MEAN ± STD OVER THREE SEEDS). * MARKS A RECALL CHANGE FROM THE SAME EXCERPTS BEFORE THE CHANGE WITH HOLM-ADJUSTED $p < 0.05$ (PER DETECTOR).

Condition	Lexical-LR	RoBERTa	Coordinator	Fast-DetectGPT	Binoculars
Writer only, no Editor	0.999±0.002	1.000±0.000	1.000±0.000	0.839±0.032*	0.855±0.041*
Paraphrase, same model	0.999±0.002	1.000±0.000	1.000±0.000	0.572±0.084*	0.548±0.074*
Paraphrase, other model	1.000±0.000	1.000±0.000	1.000±0.000	0.610±0.054*	0.575±0.066*
Paraphrase twice	1.000±0.000	1.000±0.000	1.000±0.000	0.588±0.051*	0.570±0.066*
Back-translation	0.999±0.002	0.992±0.001*	0.996±0.005	0.588±0.047*	0.495±0.040*
"Humanize" prompt	0.999±0.002	1.000±0.000	0.999±0.002	0.644±0.054*	0.668±0.046*
Advisory-style prompt	0.997±0.005	0.998±0.003	1.000±0.000	0.642±0.079*	0.637±0.077*
Few-shot human examples	0.982±0.005*	0.969±0.013*	0.950±0.013*	0.764±0.056*	0.774±0.035
Single agent reading the source	0.663±0.113*	0.721±0.052*	0.601±0.121*	0.757±0.041	0.650±0.086*
25% LLM sentences	0.303±0.050*	0.366±0.038*	0.171±0.134*	0.442±0.069*	0.258±0.057*
50% LLM sentences	0.844±0.005*	0.620±0.036*	0.535±0.023*	0.600±0.058*	0.481±0.081*
75% LLM sentences	0.976±0.004*	0.847±0.020*	0.843±0.015*	0.681±0.047*	0.637±0.066*

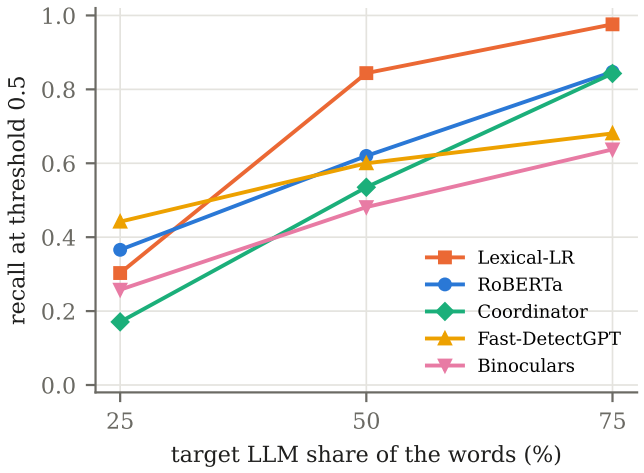


Fig. 6. Recall on partly generated texts against the share of LLM words, models trained on CISA + ABC.

and 0.269 of those that end with it, because it reads only the first 256 tokens. Lexical-LR, which reads the whole text, finds 0.862 and 0.826. The Coordinator follows RoBERTa (0.817 and 0.240).

Training on these conditions helps where the base models failed (Table VII). Adding paraphrased and single-agent texts of the training sources raises recall on single-agent test texts from 0.663 to 0.892 (Lexical-LR) and from 0.721 to 0.976 (RoBERTa), and on D from 0.801 to 0.969 and from 0.958 to 0.995, although no D text was added. The Coordinator gains less (0.601 to 0.766 on single-agent texts).

E. Calibration and decision thresholds

In-generator, calibration on validation data improves the probabilities. The expected calibration error of Lexical-LR falls from 0.167 (raw) to 0.004 with Platt scaling (Table VIII). Under generator shift the probabilities get worse. On D, the negative log-likelihood of Lexical-LR rises from 0.569 (raw) to 1.823 (Platt) and 2.037 (isotonic), and that of RoBERTa from 0.384 to 1.772 and 1.906. The calibrators learn from validation data that the scores are reliable and sharpen them, and on D the sharpened scores are wrong (Fig. 7). Temperature scaling, which can only soften or sharpen all scores together, harms RoBERTa least (NLL 0.589). The effect on decisions at 0.5 is mixed. Platt scaling moves the boundary of Lexical-LR enough to raise its F1 on D from 0.329 to 0.536, and it lowers that of RoBERTa from 0.859 to 0.750. Calibration on in-distribution data is thus neither a reliable repair for the decision nor a safe one for the probabilities.

The threshold matters more than the calibrator (Table IX, Fig. 8). Choosing the threshold with the best validation F1 is an unreliable fix, because the validation texts come from the training generator. It raises Lexical-LR on D only to 0.502 and lowers RoBERTa to 0.773. The pseudo-unseen rule, which tunes the threshold on another local generator, raises the F1 of Lexical-LR on D from 0.329 to 0.758. The two human-quantile rules, which use no LLM text at all, raise it to 0.852 (1% target) and 0.886 (5% target), and that of the Coordinator from 0.721 to 0.892 and 0.935. RoBERTa, whose D scores were already mostly above 0.5, gains less (0.859 to 0.909).

The cost shows on the other publishers. Table IX is computed for models trained on one generator, which at the fixed threshold flag only 0.003 (Lexical-LR) and 0.019 (Coordinator) of the vendor excerpts, far fewer than the pooled models of Table V. With the 5% rule these rates rise to 0.212 and 0.180. Vendor human texts score higher than CISA ones, so a quantile of CISA scores lets more of them through. Of the five rules, those that recover most of D also raise false alarms on vendor text the most, which is the trade already seen for pooled training.

F. When the Coordinator helps

Table X compares the Coordinator with RoBERTa in all 22 scenarios. RoBERTa is the reference because it is the member that loses least under shift. In-generator, Lexical-SVM is as good as the Coordinator (Table III). The Coordinator is better than RoBERTa in 13 scenarios and worse in 5 (Holm-adjusted $p < 0.05$). The gains are small where both are near the ceiling (in-generator 0.006, leave-one-out 0.010, the paraphrase conditions) and larger on vendor text (Microsoft 0.051, Mandiant 0.059), where its lower false positive rate counts. The losses are the shifts that change the evidence most: texts with a quarter of LLM words (-0.244), D after training

TABLE VII
RECALL AT THRESHOLD 0.5 (MEAN ± STD OVER THREE SEEDS), ABC TRAINING (BASE) AGAINST ABC PLUS SINGLE-AGENT AND PARAPHRASED TEXTS OF TRAINING SOURCES (AUG.).

Test	Lexical-LR		RoBERTa		Coordinator	
	base	aug.	base	aug.	base	aug.
A	0.996±0.007	0.996±0.007	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000
D	0.801±0.018	0.969±0.028	0.958±0.030	0.995±0.008	0.851±0.046	0.866±0.087
E	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000
Paraphrase, same model	0.999±0.002	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000
Paraphrase twice	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000
"Humanize" prompt	0.999±0.002	1.000±0.000	1.000±0.000	0.999±0.002	0.999±0.002	0.999±0.002
Single agent	0.663±0.113	0.892±0.051	0.721±0.052	0.976±0.009	0.601±0.121	0.766±0.119

TABLE VIII
PROBABILITY QUALITY ON TEST DATA, TRAINING ON ONE OF A, B, C (MEAN ± STD OVER THREE SEEDS). CALIBRATORS ARE FITTED ON VALIDATION DATA ONLY.

Detector	Calibrator	In-generator			D: gpt-oss-120b			E: Gemini		
		Brier	ECE	NLL	Brier	ECE	NLL	Brier	ECE	NLL
Lexical-LR	Raw	0.035±0.001	0.167±0.002	0.190±0.002	0.201±0.004	0.316±0.015	0.569±0.009	0.073±0.003	0.243±0.005	0.292±0.008
Lexical-LR	Platt	0.002±0.003	0.004±0.002	0.014±0.020	0.271±0.077	0.299±0.075	1.823±0.427	0.015±0.016	0.026±0.019	0.054±0.055
Lexical-LR	Temperature	0.003±0.004	0.006±0.005	0.025±0.033	0.353±0.048	0.380±0.023	2.899±1.656	0.030±0.008	0.050±0.012	0.135±0.050
Lexical-LR	Isotonic	0.003±0.002	0.006±0.002	0.017±0.019	0.242±0.050	0.289±0.055	2.037±0.524	0.015±0.012	0.036±0.022	0.057±0.047
RoBERTa	Raw	0.007±0.003	0.013±0.003	0.029±0.011	0.097±0.010	0.113±0.010	0.384±0.049	0.008±0.003	0.016±0.004	0.033±0.014
RoBERTa	Platt	0.008±0.005	0.010±0.005	0.059±0.046	0.170±0.098	0.181±0.096	1.772±1.409	0.020±0.016	0.024±0.015	0.202±0.197
RoBERTa	Temperature	0.007±0.002	0.014±0.005	0.050±0.022	0.102±0.010	0.114±0.008	0.589±0.062	0.009±0.003	0.017±0.005	0.055±0.019
RoBERTa	Isotonic	0.008±0.005	0.010±0.006	0.077±0.066	0.158±0.094	0.177±0.095	1.906±1.375	0.020±0.014	0.026±0.015	0.227±0.197
Coordinator	Raw	0.002±0.000	0.005±0.002	0.008±0.001	0.159±0.043	0.203±0.049	0.598±0.198	0.005±0.003	0.021±0.007	0.026±0.010
Coordinator	Platt	0.002±0.001	0.004±0.003	0.015±0.011	0.184±0.067	0.202±0.072	1.303±0.621	0.006±0.002	0.011±0.002	0.024±0.011
Coordinator	Temperature	0.002±0.000	0.003±0.002	0.015±0.012	0.191±0.058	0.202±0.053	1.916±0.854	0.005±0.002	0.009±0.003	0.035±0.022
Coordinator	Isotonic	0.002±0.000	0.004±0.002	0.014±0.014	0.173±0.055	0.208±0.059	1.492±0.684	0.006±0.003	0.015±0.007	0.030±0.020

TABLE IX
EFFECT OF THE DECISION RULE, TRAINING ON ONE OF A, B, C. EVERY THRESHOLD IS SET ON VALIDATION DATA. PSEUDO-UNSEEN USES VALIDATION TEXTS OF LOCAL GENERATORS THAT ARE NEITHER IN TRAINING NOR THE TEST GENERATOR. HUMAN 1% AND 5% USE VALIDATION HUMAN SCORES ALONE.

Detector	Threshold rule	F1, in	F1, cross	F1, D	Recall, D	F1, E	F1, ext.	FPR, ext.
Lexical-LR	Fixed 0.5	0.997±0.004	0.966±0.008	0.329±0.017	0.208	0.961±0.012	0.982±0.001	0.003
Lexical-LR	Validation F1	0.997±0.003	0.978±0.019	0.502±0.099	0.363	0.980±0.025	0.983±0.005	0.016
Lexical-LR	Pseudo-unseen F1	0.996±0.003	0.994±0.003	0.758±0.047	0.619	0.996±0.004	0.971±0.001	0.058
Lexical-LR	Human 5% FPR	0.986±0.017	0.986±0.016	0.886±0.018	0.823	0.987±0.017	0.905±0.019	0.212
Lexical-LR	Human 1% FPR	0.991±0.015	0.990±0.014	0.852±0.032	0.761	0.991±0.015	0.935±0.031	0.139
RoBERTa	Fixed 0.5	0.992±0.003	0.990±0.002	0.859±0.022	0.777	0.991±0.003	0.960±0.009	0.084
RoBERTa	Validation F1	0.992±0.006	0.982±0.005	0.773±0.171	0.685	0.979±0.015	0.969±0.011	0.057
RoBERTa	Pseudo-unseen F1	0.989±0.007	0.983±0.006	0.819±0.103	0.743	0.980±0.010	0.964±0.012	0.071
RoBERTa	Human 5% FPR	0.961±0.031	0.961±0.031	0.909±0.014	0.905	0.961±0.031	0.901±0.035	0.223
RoBERTa	Human 1% FPR	0.980±0.019	0.978±0.018	0.862±0.065	0.807	0.977±0.017	0.948±0.033	0.111
Coordinator	Fixed 0.5	0.998±0.000	0.993±0.003	0.721±0.083	0.593	0.995±0.003	0.989±0.004	0.019
Coordinator	Validation F1	0.998±0.000	0.993±0.003	0.708±0.092	0.580	0.993±0.004	0.989±0.004	0.018
Coordinator	Pseudo-unseen F1	0.992±0.008	0.991±0.003	0.803±0.087	0.705	0.992±0.008	0.980±0.013	0.041
Coordinator	Human 5% FPR	0.973±0.019	0.973±0.019	0.935±0.001	0.928	0.973±0.019	0.918±0.014	0.180
Coordinator	Human 1% FPR	0.984±0.008	0.984±0.009	0.892±0.037	0.840	0.984±0.009	0.950±0.014	0.107

on one generator (-0.138, 95% CI [-0.164, -0.113]), single-agent texts (-0.090), texts with half LLM words (-0.053) and D after training on all three (-0.043).

On validation data, the stack weights Lexical-SVM (0.538) and RoBERTa (0.526) most, then Stylometry (0.312) and Lexical-LR (0.156). The LLM Judge gets a negative weight (-0.108). On D, the mean logit of Lexical-SVM collapses, and weighted by its coefficient it accounts for 3.755 logit units of the Coordinator's drop, against 1.720 for RoBERTa and 1.484 for Stylometry. Weights fitted on the training generator thus put most trust in a member that fails on D. Selecting the single best member on validation does worse (F1 on D 0.404; the member chosen most often is Lexical-LR), the unweighted mean of logits does slightly better (0.754), and a stack fitted on validation texts of local generators absent from training does best (0.824). None reaches RoBERTa alone (0.859). Adding the two zero-shot detectors as members changes little (0.725). Removing RoBERTa from the Coordinator lowers its F1 on D to 0.490, and removing any other member changes it by at most 0.063 (Table XI).

G. Ablations

Masking agency names leaves Lexical-LR almost unchanged (0.996 in-generator, 0.967 cross), so the detectors do not depend on the word *CISA* although it tops the log-odds list. Switching off output cleaning changes nothing (0.997).

The length-matched subset keeps LLM texts within 20% of the length of their human excerpt. Generator A almost never

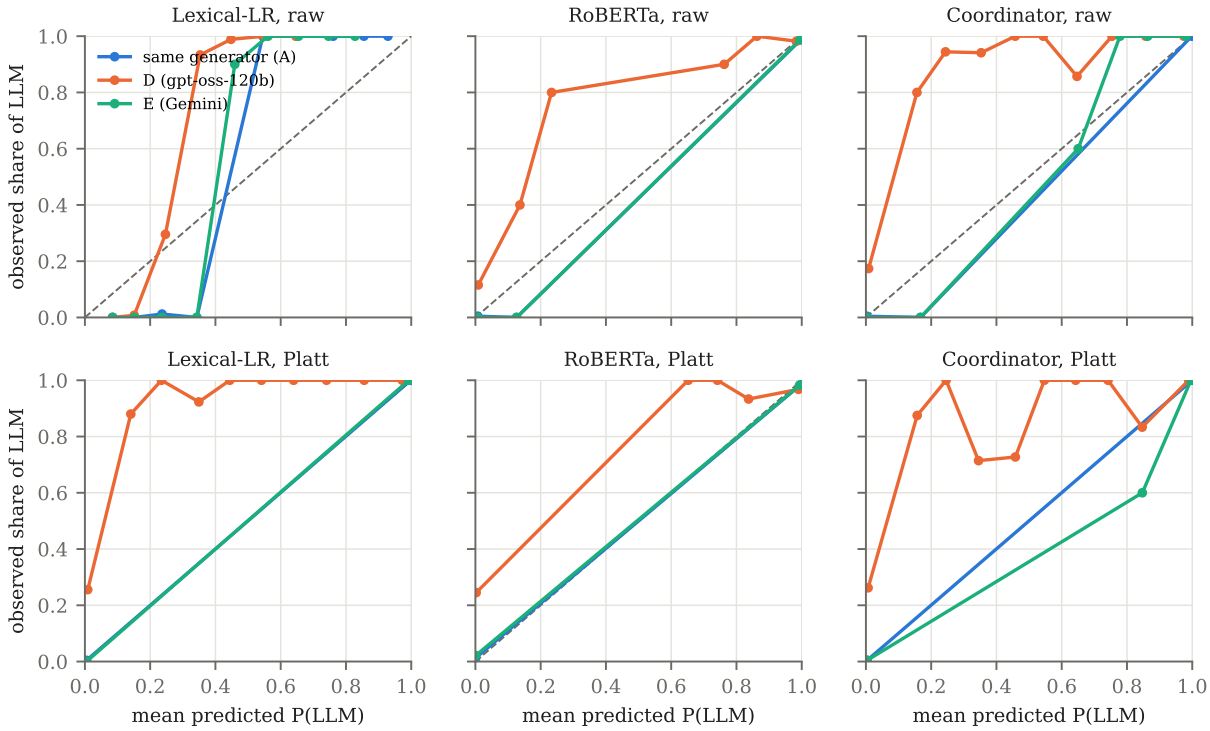


Fig. 7. Reliability of raw (top) and Platt-scaled (bottom) probabilities, models trained on A. Points are bins with at least five texts; the dashed diagonal is perfect calibration.

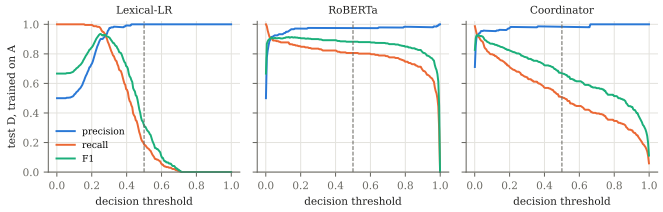


Fig. 8. Precision, recall and F1 on D against the decision threshold, models trained on A; the dashed line is 0.5.

writes that long (about 2 such texts per test split), so we report B and C, with 12 and 23 texts per split. The lexical detectors and RoBERTa are unaffected (Lexical-LR 1.000 and 1.000, RoBERTa 1.000 and 0.992). Stylometry falls from 0.917 to 0.764 on B and from 0.947 to 0.933 on C. Length therefore carries part of the stylometric signal for the short generators, most clearly for B, and these subsets are small. The feature groups give the same answer on a larger sample. Length and punctuation features reach 0.875 and 0.823 in-generator but 0.482 and 0.492 on D, which matches human length, while lexical diversity alone gives 0.863 on D, more than the full stylometric model (0.706).

Holding one Writer template out of training lowers the recall of Lexical-LR on that template to between 0.817 and 1.000, depending on the template, while its AUC stays at 1.000. No single template is needed to rank the texts, but the templates are not interchangeable at a fixed threshold. Across topics, recall of the Coordinator on the other local generators stays between 0.978 and 1.000. On D it is lowest for vulnerability and APT reports (0.523 and 0.537) and highest for the small "other" group (0.926).

H. Cost

Inference cost ranges from 1.7 ms per text for Lexical-LR to 1136.5 ms for the LLM Judge (Table XII). RoBERTa needs 2.8 ms on the GPU, Fast-DetectGPT 195.9 ms and Binoculars 725.0 ms. Run one member after another, the Coordinator costs 1168.6 ms per text, almost all of it the Judge. Without the Judge it costs 32.1 ms. The Judge's negative stacking weight and the near-zero effect of removing it mean that this time buys nothing in our data.

I. Error analysis

What the categories show. Table XIV gives the error categories with their base rates. Of the D texts that Lexical-LR misses, 76% are impersonal and of human length, but so are 76% of all D texts. The category describes how D writes and does not single out the misses. The same holds for D texts that name an agency. Two categories do depart from their base rate. D texts in the first person are missed less often than the others (Lexical-LR misses 0.638 of them against 0.799 of the impersonal ones). Texts that contain technical identifiers absent from the Analyst facts are missed more often. They are 4% of the D texts and 12% of RoBERTa's misses, and RoBERTa misses 0.633 of them against 0.212 of the impersonal ones. This group is small (30 run-level cases) and the rule is a pattern match that also fires on harmless expansions (Table XVI), so we read it as an indication: a text

TABLE X

F1 OF ROBERTA, THE COORDINATOR AND ITS VARIANTS AT THRESHOLD 0.5 IN ALL SCENARIOS (MEAN OVER RUNS; ± STD OVER SEEDS FOR THE FIRST TWO). GAP IS COORDINATOR MINUS ROBERTA, WITH A PAIRED SOURCE-LEVEL BOOTSTRAP INTERVAL; * MARKS HOLM-ADJUSTED $p < 0.05$. "UNSEEN STACK" FITS THE COORDINATOR ON VALIDATION TEXTS OF GENERATORS ABSENT FROM TRAINING. "MULTI-AGENT, SAME SOURCES" IS THE REFERENCE FOR THE SINGLE-AGENT ROW.

Scenario	RoBERTa	Coord.	Gap	95% CI	Best on val.	Mean logit	Unseen stack	+ zero-shot
In-generator	0.992±0.003	0.998±0.000	+0.006*	[+0.002, +0.010]	0.996	0.998	0.994	0.998
Cross, local	0.990±0.002	0.993±0.003	+0.003	[-0.001, +0.007]	0.974	0.994	0.994	0.993
Leave-one-out	0.984±0.002	0.994±0.001	+0.010*	[+0.005, +0.017]	0.996	0.996	–	0.995
D, one generator	0.859±0.022	0.721±0.083	-0.138*	[-0.164, -0.113]	0.404	0.754	0.824	0.725
D, ABC	0.958±0.012	0.915±0.030	-0.043*	[-0.072, -0.018]	0.788	0.960	–	0.928
E, one generator	0.991±0.003	0.995±0.003	+0.004	[-0.001, +0.009]	0.966	0.995	0.994	0.993
E, ABC	0.979±0.003	0.996±0.004	+0.016*	[+0.006, +0.028]	0.987	0.998	–	0.996
Microsoft, ABC	0.927±0.012	0.978±0.022	+0.051*	[+0.046, +0.057]	0.945	0.962	–	0.981
Mandiant, ABC	0.910±0.017	0.969±0.007	+0.059*	[+0.051, +0.066]	0.949	0.945	–	0.974
Writer only, ABC	0.979±0.003	0.996±0.004	+0.016*	[+0.010, +0.023]	0.987	0.998	–	0.996
Paraphrase, ABC	0.979±0.003	0.996±0.004	+0.016*	[+0.010, +0.023]	0.986	0.998	–	0.996
Paraphrase, other model, ABC	0.979±0.003	0.996±0.004	+0.016*	[+0.010, +0.023]	0.986	0.998	–	0.996
Paraphrase twice, ABC	0.979±0.003	0.996±0.004	+0.016*	[+0.011, +0.023]	0.986	0.998	–	0.996
Back-translation, ABC	0.975±0.003	0.994±0.006	+0.019*	[+0.012, +0.026]	0.985	0.996	–	0.993
Humanize, ABC	0.979±0.003	0.995±0.002	+0.016*	[+0.010, +0.022]	0.987	0.998	–	0.995
Advisory-style prompt, ABC	0.979±0.003	0.996±0.004	+0.017*	[+0.011, +0.023]	0.985	0.998	–	0.996
Few-shot prompt, ABC	0.963±0.003	0.970±0.010	+0.007	[-0.002, +0.015]	0.961	0.980	–	0.974
Multi-agent, same sources, ABC	0.994±0.010	1.000±0.000	+0.006	[+0.000, +0.012]	0.994	1.000	–	1.000
Single-agent, ABC	0.812±0.045	0.722±0.107	-0.090*	[-0.138, -0.049]	0.688	0.784	–	0.730
25% LLM, ABC	0.518±0.041	0.274±0.184	-0.244*	[-0.278, -0.210]	0.370	0.333	–	0.263
50% LLM, ABC	0.745±0.025	0.691±0.023	-0.053*	[-0.074, -0.035]	0.725	0.758	–	0.686
75% LLM, ABC	0.896±0.014	0.910±0.007	+0.014*	[+0.004, +0.024]	0.922	0.938	–	0.910

TABLE XI

ABLATIONS, F1 (MEAN ± STD OVER SEEDS; TRAINING ON ONE OF A, B, C).

Condition	In-generator	Cross (A–C)	D	E
TF-IDF only (LR)	0.997 ± 0.004	0.966 ± 0.008	0.329 ± 0.017	0.961 ± 0.012
Stylometry only (XGB)	0.934 ± 0.006	0.868 ± 0.004	0.706 ± 0.013	0.778 ± 0.022
TF-IDF + stylometry (LR)	0.964 ± 0.005	0.919 ± 0.010	0.735 ± 0.060	0.839 ± 0.021
LR, no output cleaning	0.997 ± 0.004	0.965 ± 0.008	0.336 ± 0.018	0.955 ± 0.012
LR, agency names masked	0.996 ± 0.004	0.967 ± 0.008	0.349 ± 0.016	0.964 ± 0.012
RoBERTa, agency names masked	0.988 ± 0.004	0.986 ± 0.002	0.901 ± 0.035	0.985 ± 0.007
Stylometry: length only	0.875 ± 0.008	0.790 ± 0.004	0.482 ± 0.060	0.657 ± 0.039
Stylometry: lexical only	0.845 ± 0.025	0.816 ± 0.024	0.863 ± 0.025	0.865 ± 0.033
Stylometry: punctuation only	0.823 ± 0.012	0.771 ± 0.024	0.492 ± 0.055	0.590 ± 0.043
Stylometry: repetition only	0.800 ± 0.020	0.769 ± 0.024	0.825 ± 0.020	0.775 ± 0.037
Stylometry: POS only	0.814 ± 0.038	0.755 ± 0.036	0.777 ± 0.052	0.850 ± 0.063
Coordinator, all agents	0.998 ± 0.000	0.993 ± 0.003	0.721 ± 0.083	0.995 ± 0.003
Coordinator without Lexical-LR	0.998 ± 0.000	0.993 ± 0.003	0.727 ± 0.084	0.995 ± 0.003
Coordinator without Lexical-SVM	0.996 ± 0.002	0.990 ± 0.008	0.784 ± 0.058	0.991 ± 0.002
Coordinator without LLM Judge	0.999 ± 0.001	0.993 ± 0.004	0.732 ± 0.102	0.996 ± 0.004
Coordinator without Stylometry	0.998 ± 0.002	0.994 ± 0.001	0.722 ± 0.125	0.992 ± 0.004
Coordinator without RoBERTa	0.997 ± 0.003	0.972 ± 0.007	0.490 ± 0.094	0.960 ± 0.006

that spells out file names and identifiers looks more like a human advisory to these detectors. For the other generators the few misses concentrate on texts that name an agency (7% of the texts, 38% of RoBERTa's misses).

Missed against caught D texts. Table XV compares the D texts that at least half of the runs miss with the rest (152 against 39 texts for Lexical-LR, 35 against 156 for RoBERTa). Missed texts are longer ($\delta = 0.318$ for Lexical-LR, 0.250 for RoBERTa, 0.520 for the Coordinator) and less diverse (type-token ratio $\delta = -0.389$, -0.528 and -0.656). For Lexical-LR and the Coordinator they also contain more proper nouns ($\delta = 0.356$ and 0.346), which is not the case for RoBERTa (-0.019). The largest single contrast for Lexical-LR is the cue rate. Missed texts carry fewer of the n-grams the model associates with LLM text ($\delta = -0.504$) and more of those it associates with human text ($\delta = 0.276$). In the regression, one standard deviation more length multiplies the odds of a miss by 1.540 for Lexical-LR (95% CI [1.187, 1.999]), 2.070 for RoBERTa and 2.624 for the Coordinator. A higher proper-noun share raises them for Lexical-LR (1.723) and the Coordinator (1.550) but not clearly for RoBERTa (1.136, [0.892, 1.447]). Higher MTLD lowers the odds for all three (0.637, 0.532, 0.624). The top row of Fig. 9 shows the same relations as miss rates by tertile.

Why D is missed and E is not. D and E are both large models and both match the human length, so length does not explain the difference between them. Table XIII points to voice and vocabulary. First-person pronouns occur in 10% of the D texts, fewer than in the CISA excerpts (14%), and in 45% of the E texts, which is the range of the local generators (39%

TABLE XII
COST AND PERFORMANCE, ONE TRAINING GENERATOR. INFERENCE ON ONE RTX 4060 LAPTOP GPU, THREE REPETITIONS OVER A FIXED SAMPLE. THE COORDINATOR COST IS THE SUM OF ITS MEMBERS RUN ONE AFTER ANOTHER; "FIT" EXCLUDES THE LLM JUDGE, WHICH IS NOT TRAINED.

Detector	ms/text	Rel.	GPU MiB	Fit s	F1, in	F1, D	AUC, D	F1, E
Lexical-LR	1.7	0.60	–	1	0.997	0.329	0.979	0.961
Lexical-SVM	2.0	0.72	–	1	0.998	0.327	0.983	0.969
Stylometry	25.6	9.27	–	0	0.934	0.706	0.878	0.778
RoBERTa	2.8	1.00	1269	20	0.992	0.859	0.982	0.991
LLM Judge	1136.5	410.75	16	–	0.504	0.499	0.533	0.628
Fast-DetectGPT	195.9	70.78	5771	–	0.744	0.540	0.699	0.427
Binoculars	725.0	262.02	5005	–	0.817	0.365	0.734	0.307
Coordinator	1168.6	422.34	–	–	0.998	0.721	0.987	0.995

TABLE XIII
MEAN TEXT PROPERTIES BY SOURCE (MULTI-AGENT TEXTS FOR A–E). FIRST PERSON AND AGENCY: SHARE OF TEXTS WITH AT LEAST ONE FIRST-PERSON PRONOUN OR AGENCY NAME. CUES: OCCURRENCES PER 100 WORDS OF THE WORD N-GRAMS WITH THE LARGEST LEXICAL-LR WEIGHTS ON EACH SIDE (MODEL TRAINED ON A).

Source	Words	MTLD	Proper nouns	First person	Agency named	LLM-side cues	Human-side cues
Human (CISA)	279	86	0.118	14%	78%	0.85	6.16
A: Llama 3.1 8B	161	94	0.058	47%	10%	7.19	2.22
B: Qwen 2.5 7B	166	125	0.064	39%	11%	5.57	2.59
C: Mistral 7B	210	134	0.064	61%	17%	5.19	2.92
D: gpt-oss-120b	294	206	0.051	10%	13%	2.15	3.32
E: Gemini 3.1 Flash-Lite	279	170	0.035	45%	9%	3.60	2.27
Human (Microsoft)	312	95	0.063	77%	1%	1.43	3.69
Human (Mandiant)	295	93	0.078	74%	4%	1.77	3.57

TABLE XIV
ERROR CATEGORIES OVER ALL SEEDS: COUNT AND SHARE OF THE ERRORS OF EACH DETECTOR. CATEGORIES ARE MUTUALLY EXCLUSIVE AND ASSIGNED BY THE FIRST RULE THAT APPLIES (SECTION V-E); A TEXT COUNTS ONCE PER RUN IN WHICH IT IS MISCLASSIFIED. "ALL TEXTS" IS THE SHARE OF THE CATEGORY AMONG ALL TEXTS OF THE GROUP, ERRORS OR NOT, SO A CATEGORY IS OVER-REPRESENTED AMONG THE ERRORS ONLY WHEN ITS SHARE EXCEEDS THIS. FN AND CISA ROWS: MODELS TRAINED ON ONE GENERATOR; VENDOR ROWS: MODELS TRAINED ON ABC.

Error	Category	All texts	Lexical-LR	RoBERTa	Coordinator
FN, D	impersonal, human length	76%	429 (76%)	114 (72%)	211 (74%)
FN, D	cites an agency	11%	60 (11%)	15 (9%)	28 (10%)
FN, D	first person, still missed	10%	44 (8%)	10 (6%)	22 (8%)
FN, D	unsupported technical detail	4%	30 (5%)	19 (12%)	26 (9%)
FN, other generators	first person, still missed	46%	44 (30%)	6 (46%)	11 (50%)
FN, other generators	impersonal, short	30%	43 (30%)	1 (8%)	2 (9%)
FN, other generators	cites an agency	7%	28 (19%)	5 (38%)	6 (27%)
FN, other generators	impersonal, human length	17%	29 (20%)	1 (8%)	3 (14%)
FN, other generators	unsupported technical detail	0%	1 (1%)	0	0
FP, CISA	first person	15%	0	3 (30%)	1 (50%)
FP, CISA	product vulnerability notice	9%	0	3 (30%)	0
FP, CISA	few proper nouns	19%	0	3 (30%)	0
FP, CISA	other	28%	0	1 (10%)	1 (50%)
FP, vendor blogs	first person	75%	170 (89%)	212 (91%)	61 (85%)
FP, vendor blogs	few proper nouns	16%	20 (11%)	19 (8%)	11 (15%)
FP, vendor blogs	product vulnerability notice	2%	0	2 (1%)	0

to 61%). The n-grams that Lexical-LR weights towards LLM occur 2.146 times per 100 words in D, 3.596 in E and between 5.189 and 7.187 in the local generators (CISA: 0.852). D also uses more of the human-side n-grams than any other generator (3.319). E writes at human length but keeps the *we* and *our* of the smaller models and their advice vocabulary, and a detector trained on those models recognizes it. D writes impersonally, repeats identifiers and does not address the reader. What still marks it as LLM text is its lexical diversity (median MTLD 199 against 84 for CISA), which is why the lexical-diversity features alone detect D better than the full stylometric model and why the least diverse D texts are the ones missed. The Writer template matters as well. Lexical-LR misses 0.917 of the D texts written from the incident-report template and 0.671 of those from the SOC-bulletin template, and RoBERTa 0.305 from the vendor-blog template against 0.114. The sampling temperature does not (0.776 at 0.7 and 0.812 at 1.0 for Lexical-LR).

False positives. On CISA test excerpts Lexical-LR makes no false positive and RoBERTa makes 10 over all runs: 3 excerpts written in the first person, 3 product vulnerability notices, 3 with few proper nouns and 1 other. These counts are too small for more than a description. The vendor excerpts give enough cases (190 run-level false positives for Lexical-LR, 233 for RoBERTa, 72 for the Coordinator). Vendor blogs write in the first person as a rule (77% of Microsoft and 74%

TABLE XV
PROPERTIES OF MISCLASSIFIED AGAINST CORRECTLY CLASSIFIED TEXTS: MEANS OF THE TWO GROUPS AND CLIFF'S δ (POSITIVE WHEN THE MISCLASSIFIED TEXTS HAVE MORE). A TEXT BELONGS TO THE FIRST GROUP WHEN AT LEAST HALF OF THE RUNS MISCLASSIFY IT. * MARKS A MANN-WHITNEY TEST WITH BENJAMINI-HOCHBERG ADJUSTED $p < 0.05$. D: MODELS TRAINED ON ONE GENERATOR; VENDOR EXCERPTS: MODELS TRAINED ON ABC.

Group	Property	Lexical-LR		RoBERTa		Coordinator	
		means	δ	means	δ	means	δ
D texts, missed / caught	Texts	152 / 39		35 / 156		90 / 101	
	Words	301 / 258	+0.32*	322 / 285	+0.25*	331 / 258	+0.52*
	MTLD	203 / 220	-0.25*	183 / 212	-0.37*	195 / 217	-0.34*
	Type-token ratio	0.680 / 0.718	-0.39*	0.647 / 0.696	-0.53*	0.654 / 0.717	-0.66*
	Proper-noun share	0.055 / 0.038	+0.36*	0.052 / 0.052	-0.02	0.061 / 0.043	+0.35*
	Uses first person (share of texts)	0.08 / 0.18	-0.10	0.06 / 0.11	-0.05	0.09 / 0.11	-0.02
	LLM-side cues per 100 words	2.01 / 3.16	-0.50*	1.69 / 2.37	-0.33*	1.91 / 2.55	-0.29*
	Human-side cues per 100 words	3.43 / 2.85	+0.28*	3.85 / 3.19	+0.32*	3.47 / 3.17	+0.15
	Identifiers not in the facts	0.07 / 0.00	+0.05	0.17 / 0.03	+0.09*	0.11 / 0.00	+0.09*
Vendor excerpts, flagged / passed	Texts	64 / 374		79 / 359		15 / 423	
	Words	282 / 307	-0.32*	297 / 305	-0.13	249 / 306	-0.48*
	MTLD	108 / 92	+0.39*	109 / 91	+0.43*	121 / 93	+0.64*
	Type-token ratio	0.587 / 0.556	+0.42*	0.588 / 0.554	+0.38*	0.635 / 0.558	+0.76*
	Proper-noun share	0.071 / 0.070	+0.01	0.060 / 0.072	-0.22*	0.060 / 0.071	-0.25
	Pronoun share	0.035 / 0.035	+0.03	0.036 / 0.035	+0.09	0.039 / 0.035	+0.21
	First person per 100 words	1.47 / 0.86	+0.35*	1.44 / 0.84	+0.38*	1.19 / 0.94	+0.14
	LLM-side cues per 100 words	2.44 / 1.44	+0.45*	2.31 / 1.43	+0.43*	2.16 / 1.57	+0.30
	Human-side cues per 100 words	3.09 / 3.73	-0.28*	3.20 / 3.73	-0.25*	3.26 / 3.65	-0.20

of Mandiant excerpts against 14% for CISA), and the first-person category is over-represented among the false alarms (89% for Lexical-LR and 91% for RoBERTa against a base rate of 75%). Table XV and the bottom row of Fig. 9 show which excerpts are flagged. They are more diverse (MTLD $\delta = 0.389$, 0.432 and 0.637), use the first person more often per 100 words (0.354 and 0.382), carry more LLM-side cues (0.447 and 0.432) and, for Lexical-LR and the Coordinator, are shorter (-0.325 and -0.480). The overall pronoun share, which separates vendors from CISA at the corpus level, does not separate flagged from passed vendor excerpts (0.028 and 0.092), and the proper-noun share does so only for RoBERTa (-0.220). In the regression, one standard deviation more first-person use multiplies the odds of a false alarm by 1.595 for Lexical-LR (95% CI [1.237, 2.058]) and 1.754 for RoBERTa, and one standard deviation more MTLD by 1.964, 1.987 and 2.634 for the Coordinator.

Examples. Table XVI shows six confident errors. The first D text is counted as adding an identifier because it writes `certutil.exe` where the facts say CertUtil, which shows how coarse that rule is. The second expands the acronym SVR as "Federal Security Service", the name of a different Russian agency (the FSB), an error of content that a style detector cannot see. The two CISA false positives are a joint advisory written as *we* and a narrative summary with few names.

Taken together, the two kinds of error are mirror images. The LLM texts that are missed are long, impersonal, rich in proper nouns and low in lexical diversity, the profile of a CISA excerpt. The human texts that are flagged are shorter, written in the first person and lexically diverse, the profile of the training generators.

VII. DISCUSSION

A. Answers to the research questions

Q1, linguistic differences. LLM CTI has higher lexical diversity, repeats fewer bigrams, names fewer entities and gives more advice to the reader. Human CTI repeats identifiers and cites other documents. This agrees with the GLTR observation that generated text avoids the long tail of specific tokens [12]. Not all of these features describe human authorship as such. First-person voice separates CISA from the small generators and from the vendor blogs alike, so it describes CISA's house style. Lexical diversity is the more general difference. The three human publishers lie close together on MTLD and generators B to E lie well above them, while A, the least diverse generator, overlaps with the vendor blogs (Table XIII).

Q2, baselines against the transformer. In-generator, a TF-IDF baseline is as good as RoBERTa. The difference appears under shift. On D, RoBERTa keeps an F1 of 0.859 against 0.329, and it does not need agency names to do so (0.901 with names masked). Pre-trained representations seem to capture properties of LLM prose beyond the n-grams of the training generators, as the supervised RoBERTa detectors for GPT-2 did [14]. The training-free detectors, which need no CTI data, rank texts far worse than any trained model here. With a 2.7B scoring model and 4-bit 7B models they are weaker than in their authors' settings, and they are also 70.8 and 262.0 times slower than RoBERTa.

Q3 and Q4, generator style and cross-generator testing. Among the three 7–8B local models, cross-generator F1 stays near the in-generator value, so the detectors do not learn the style of one model only. Part of what they learn is specific to how small models write CTI: short, in the first person, with advice to the reader. Gemini writes longer but keeps that voice and is detected. gpt-oss-120b drops it and is missed by the lexical models, whose AUC nevertheless stays high. This is the pattern reported by M4 and RAID, where detectors label text from unseen generators as human [4], [5]. Our threshold results locate it in the decision, and the error analysis names the properties involved.

Q5, important features. The stylometric model splits mostly on length (top feature: n_words) and proper-noun share, the

TABLE XVI
EXAMPLES OF CONFIDENT ERRORS (CISA EXCERPTS AND LLM OUTPUT), WITH THE GENERATOR, THE LENGTH AND THE MEAN SCORE OVER THE RUNS THAT MISCLASSIFY THE TEXT.

Error	Opening of the text
FN, Coordinator: unsupported technical detail. Generator D, 341 words, mean P(LLM) 0.00	Malicious cyber actors have been observed deploying the KONNI malware family in campaigns that rely on a classic spear-phishing attachment workflow (ATT&CK T1566.001). The initial infection vector typically consists of a malicious Microsoft Office document that contains a Visual Basic for Applications (VBA) macro. When the recipient enables macros, the VBA code executes a chain of actions designed to establish…
FN, Coordinator: impersonal, human length. Generator D, 345 words, mean P(LLM) 0.07	The investigation began after the security operations center received an alert that an administrative account in the organization's cloud-based email environment had been used to modify permissions and access a number of privileged mailboxes. Log correlation and forensic analysis quickly revealed activity consistent with tactics employed by Russian Federal Security Service (SVR) cyber-espionage groups. The adversary's foothold was established through…
FN, Lexical-LR: cites an agency. Generator D, 317 words, mean P(LLM) 0.34	Malicious cyber actors-most likely advanced persistent threat (APT) groups-have been observed exploiting a series of critical VMware vulnerabilities disclosed in early 2022, specifically CVE-2022-22954, CVE-2022-22960, CVE-2022-22972, and CVE-2022-22973. These flaws enable a server-side template injection (SSTI) that can be leveraged to achieve remote code execution (RCE) on affected VMware products and, in many cases, to elevate privileges to the root…
FN, RoBERTa: first person, still missed. Generator A, 122 words, mean P(LLM) 0.15	Malicious cyber actors, likely advanced persistent threat (APT) actors, have been exploiting critical vulnerabilities affecting VMware products. The vulnerabilities, specifically CVE-2022-22954 and CVE-2022-22960, were addressed by VMware with updates on April 6, 2022. The US Cybersecurity and Infrastructure Security Agency (CISA) added these vulnerabilities to its catalog of Known Exploited Vulnerabilities on April 14 and 15, 2022, respectively. Federal agencies…
FP, RoBERTa: first person. CISA, 333 words, mean P(LLM) 0.61	Since early 2018, we have observed an increase in JBiFrost RAT being used in targeted attacks against critical national infrastructure owners and their supply chain operators. There has also been an increase in the RAT's hosting on infrastructure located in our countries. In early 2017, Adwind RAT was deployed via spoofed emails designed to look as if they originated from…
FP, RoBERTa: few proper nouns. CISA, 309 words, mean P(LLM) 0.80	The BeagleBoyz use a variety of tools and techniques to gain access to a financial institution's network, learn the topology to discover key systems, and monetize their access. The technical analysis below represents an amalgamation of multiple known incidents, rather than details of a single operation. These findings are presented to highlight the group's ability to tailor their techniques to…

lexical model on pronouns, hedging words and agency names. The length cue does not transfer to larger generators. The lexical-diversity cue does, and within D the misses are the long texts with low diversity and many proper nouns.

Q6, topics and attack types. Among local generators recall does not depend on the topic. On D, vulnerability and APT reports are missed more often than phishing and ransomware reports (0.523 and 0.537 against 0.667 and 0.633). A possible reason is that the facts of such reports are mostly actor, product and CVE names, which raise the proper-noun share of the generated text. The topic labels come from keyword rules and the standard deviations across runs are large, so we read this as a tendency.

Q7, changes in LLM technology. The detectors trained on 2024-era 7–8B models lose most on the larger and newer gpt-oss-120b, and they fail on texts produced by a different procedure with the same models, single-agent rewriting and sentence-level mixing. Paraphrasing affected the two groups of detectors differently. It lowered the recall of the two likelihood-based detectors, as in general-domain studies [7], [8], and left that of the trained ones unchanged. A plausible reason is that our paraphrasers are the same small instruction-tuned models. A paraphrase is then another LLM text with the same habits, which a classifier trained on such texts recognizes even when its token probabilities change. A paraphraser trained to evade detection, as in RADAR [18], would be a stronger test.

B. Ranking and decision

Across generator and publisher shift, the supervised detectors kept an AUC near 1 while F1 or the false positive rate moved. A detector that meets new text can usually still order it, and what fails is the boundary fitted to the training generators and the training publisher. Calibrating on in-distribution validation data does not repair this, in line with Ovadia et al. [27]. The two repairs that do work for an unseen generator, a threshold from human scores and training on several generators, both move the boundary towards human text, and both are paid for on a publisher the model has not seen. The false positive rate of Lexical-LR on Microsoft excerpts goes from 0.003 to 0.152 when it is trained on three generators instead of one. The error analysis says why the two shifts pull in opposite directions. The detectors separate CISA from the small generators along length, first-person voice and lexical diversity. An unseen generator that sits near CISA on the first two is missed, and an unseen publisher that sits near the generators on the last two is flagged. A single boundary on these properties cannot serve both. The remedy that did not have this cost was human text from more than one publisher. Adding the excerpts of one vendor to training, 231 or 207 of them with their LLM versions, cut the false positive rate on the other vendor several fold.

C. When combining agents helps

The stacked Coordinator helps when its members fail in different places, as on vendor text, where its false positive rate is less than half that of Lexical-LR or RoBERTa. It falls behind RoBERTa when the member it trusts most is the one that fails, as on D after single-generator training. The weights are fitted where all members look reliable, so the stack cannot know which member will transfer. Learning the weights under

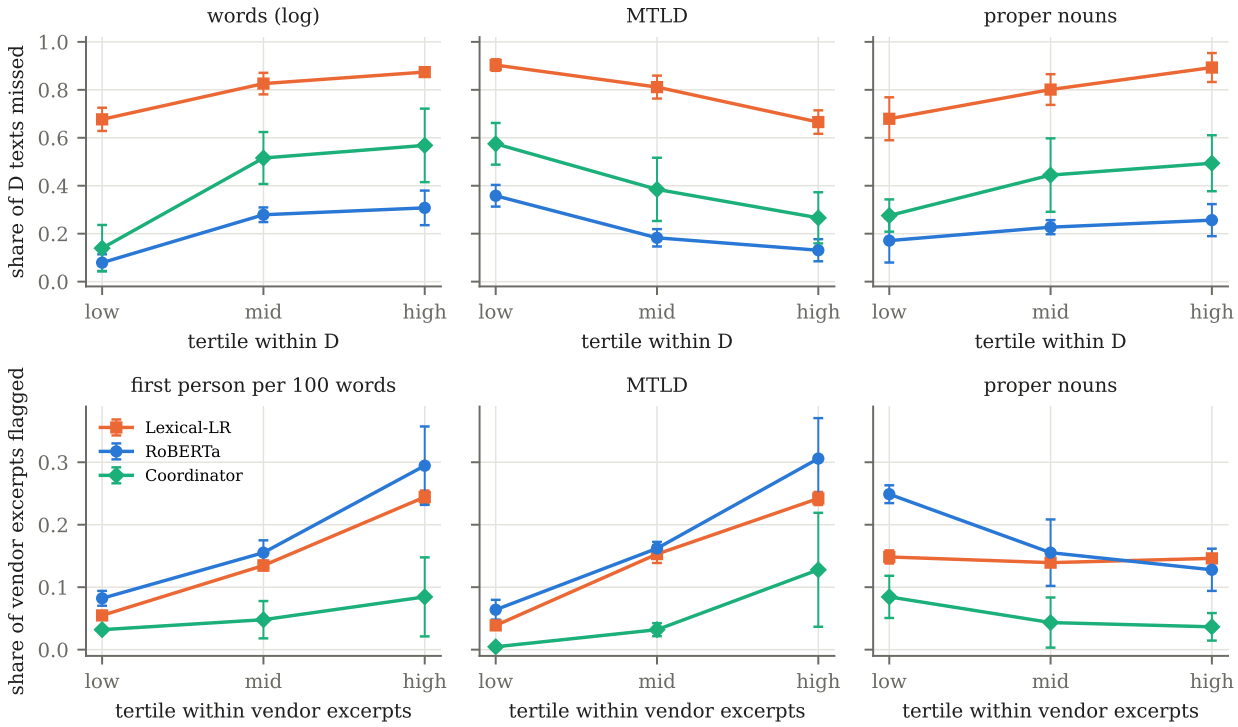


Fig. 9. Error rate by tertile of a text property (mean and standard deviation over seeds). Top: share of D texts missed by models trained on one generator. Bottom: share of Microsoft and Mandiant human excerpts flagged by models trained on ABC.

a simulated shift recovers part of the loss, which supports this reading. For a deployment, this suggests choosing the combination rule on held-out generators and publishers and dropping members, like the LLM Judge here, that add cost and no signal.

D. Use in a CTI workflow

For triage of incoming reports, a fine-tuned encoder is the best single choice in our data. On a GPU it is about as cheap as a TF-IDF model at inference, it keeps most of its recall under generator shift and it is not fooled by LLM paraphrasing. Its weaknesses are partial generation, where it sees only the first 256 tokens, and false alarms on unfamiliar publishers. Its score should therefore send a report to an analyst and not block it, long reports should be scored in windows, and human text from each new source of reports should be added to training before the detector is trusted on it. Fabricated reports that copy names and identifiers from a real report, as our single-agent texts do, and reports that mix human and generated paragraphs remain the largest gap.

VIII. Limitations and Threats to Validity

A. Dataset

Training publisher. All training text comes from CISA, so "human" in training also means "this agency's house style". The Microsoft and Mandiant sets measure how far that bias carries, but they are vendor blogs selected by publisher labels and title rules, and 80 plus 80 sources are a small sample of vendor writing. Other forms of CTI, such as STIX objects, analyst chat or non-English reports, are not covered. *Size.* 203 CISA sources give test splits of about 40 sources per seed, which is why we report source-level intervals next to the means. Several subsets are much smaller: the single-agent test texts, the length-matched subsets and the error categories with a low base rate. *Dates.* Human authorship rests on publication dates on or before 30 November 2022. CISA pages show revision dates, Microsoft metadata were overwritten by a site migration and Mandiant posts were moved to a new host, so we date documents by advisory codes, URL dates and visible bylines and remove paragraphs that announce later updates. A paragraph revised after the cutoff without such a note would go unnoticed. *Excluded pages.* 6 CISA analysis reports could not be read because the local antivirus flagged embedded malware samples.

B. Generation

Generator coverage. The three training generators are 7–8B open models from 2024. D and E exist only in the multi-agent condition and, apart from texts generated before the quota restriction, only for test excerpts, so we cannot train on larger models or test single-agent rewriting and external publishers with them. The contrast between D and E rests on one model of each kind. *One pipeline.* All LLM texts come from one set of prompts and one Analyst. The single-agent and prompt-strategy conditions show that another procedure changes the results. *Fact drift.* Generators sometimes changed or added facts. Such texts were not filtered, and a reader with domain knowledge could detect some of them by content. Our measure of added identifiers is a pattern match and misses invented

narrative. *No human-edited texts.* An attacker could have a person revise an LLM draft. We did not collect such texts, and we chose not to simulate them with another LLM, because that would measure LLM rewriting again. The sentence-level mixtures approximate a report assembled from human and machine paragraphs, not a human edit of machine text, and their LLM label is a convention.

C. Detectors and analysis

Model choice. RoBERTa is the only fine-tuned encoder and Gemma 2 9B the only judge, and no hyperparameter search was done. Binoculars ran in 4-bit precision with a truncated observer distribution, which shifts its scores, and Fast-DetectGPT with a single 2.7B model. Both are smaller than the strongest settings of their authors, and Ghostbuster could not be run. *Calibration data.* Calibrators and thresholds come from validation splits of 20 or so sources per seed. A 99th percentile of some forty human scores is close to their maximum, and the 1% rule realizes false positive rates between 0.018 and 0.040 on CISA test text. *Error analysis.* The categories are rule-based, defined by one author and not checked by a second annotator. The property comparisons and regressions are observational. They show which texts are misclassified and do not prove that changing a property would change the decision. The cue lists come from one model trained on A. *Statistics.* Bootstrap intervals resample test sources but treat the three seeds and the training generators as given, so they do not capture the variance of a different choice of generators. Their p-values cannot be smaller than $2/2001$.

IX. Conclusion

Human and LLM-written CTI differ in measurable ways, and when the generator is known a TF-IDF model is enough to separate them. The difficulty starts with text the detector has not seen. In our corpus, where five generators write from the same extracted facts, the detectors kept ranking such text correctly and lost the decision: a large unseen generator that writes impersonally was scored below the threshold, and vendor blogs written in the first person were scored above it. Pooling training generators or setting the threshold from human scores repaired the first failure and aggravated the second, and calibration on validation data repaired neither. Human text from a second publisher in training was the one remedy without that cost. A stacked combination of detector agents was worth its cost on vendor text and fell behind RoBERTa where its most trusted member failed. Rewriting an LLM text with an LLM did not hide it from trained detectors, but rewriting from the source and mixing human with generated sentences did. The study is limited to one government publisher for training, three small training generators and one generation pipeline. Human-edited LLM drafts, larger training generators and windowed scoring of long reports are the next steps.

Reproducibility

Code, configuration files, the CISA corpus, all generated texts with every agent message, predictions and result tables are in the project repository.[1] Every stage runs from one command, `python -m src.cli <stage> --config <file>`, and `experiments/run_all.sh` runs them in order. Model names, prompts, sampling temperatures, seeds and detector settings are in `configs/`, and package versions are pinned in `requirements.txt`. Every number in this paper is written by the report stage into a macro file, so the text cannot drift from the results. The Microsoft and Mandiant excerpts are rebuilt by the collection script from the listed URLs and checked against stored hashes. API generations (D, E) cannot be repeated exactly, so their outputs are stored.

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